

Link Invariants from Drinfeld Doubles of Finite Groups

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Abstract

In this thesis, we begin by introducing the necessary concepts from category theory, including k -linear additive categories, abelian categories, monoidal categories, braided tensor categories, and ribbon categories. We also present the graphical calculus of ribbon diagrams used to represent distinguished morphisms in ribbon categories.

We then study the Drinfeld doubles of finite cyclic groups and derive a general formula for the corresponding unnormalized S -matrix associated with the Hopf link. We extend this formula to arbitrary finite abelian groups and compute the invariant of a more complicated two-component link. More generally, we obtain a formula for links with an arbitrary number of components coloured by representations associated with a finite abelian group.

We subsequently turn to the Drinfeld double of an arbitrary finite group. Using the classification of its representations, we determine their duals and derive formulas for the action of the universal R -matrix on different choices of representations. These results yield a general expression for the unnormalized S -matrix of the Hopf link. As a concrete non-abelian example, we apply this expression to compute the S -matrix for S_3 .

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CHAPTER 1

Introduction

One of the central problems in knot theory is deciding whether two knots are equivalent. Since answering this question directly is often difficult, one instead searches for invariants that help distinguish knots, braids, links, tangles, and framed tangles. This project is a small exploration of some of these invariants and the mathematics behind them.

A monoidal category is a category $\mathcal{C}$ equipped with a bifunctor $\otimes$, an associativity isomorphism α , a unit object $\mathbf{1}$, and unit isomorphisms λ and ρ satisfying the usual coherence axioms. Furthermore, suppose that $\mathcal{C}$ is equipped with a natural isomorphism

$$\sigma_{a,b} : a \otimes b \longrightarrow b \otimes a.$$

Given objects $V_1, \dots, V_n$, the morphism

$$\sigma_{V_i, V_{i+1}} : V_1 \otimes \dots \otimes V_i \otimes V_{i+1} \otimes \dots \otimes V_n \longrightarrow V_1 \otimes \dots \otimes V_{i+1} \otimes V_i \otimes \dots \otimes V_n$$

is represented by the elementary braid b_i , while the associativity and unit isomorphisms correspond to identity braids. More generally, every morphism obtained by composing α , λ , ρ , σ , and their inverses can be represented by a braid diagram, where composition of maps corresponds to the braid group operation.

The coherence theorem for braided tensor categories states that two such morphisms are equal if they are represented by the same braid. Thus, instead of discussing braids, we may discuss such morphisms, and vice versa.

However, braids have free ends, whereas knots and links are obtained by joining the ends of strands. To describe this operation categorically, we need a way to bend strands. This motivates the notion of a rigid tensor category. In a rigid tensor category, every object admits a dual together with evaluation and coevaluation morphisms. Diagrammatically, these morphisms are represented by cups and caps, allowing strands to bend and enabling the representation of knots, links, and tangles.

Even with duality, one piece of topological information remains missing. When working with ribbons instead of one-dimensional strands, a ribbon may carry a full twist, and this twist cannot, in general, be removed by an isotopy. To keep track of this additional information, every object X is equipped with a natural isomorphism

$$\theta_X : X \longrightarrow X,$$

called the *twist*, satisfying suitable compatibility conditions with the braiding and duality. A braided rigid tensor category together with such a family of twist morphisms is called a ribbon category.

The graphical calculus of ribbon categories provides an algebraic model for framed knots, links, and tangles. This correspondence between topology and category theory is one of the fundamental ideas behind the construction of many quantum invariants, which form the main theme of this project.

The main aim of this thesis is to study link invariants arising from the representation categories of Drinfeld doubles of finite groups. These representation categories provide natural

examples of braided and ribbon categories, and their universal R -matrices determine the braiding used in constructing link invariants.

We begin with the Drinfeld doubles of finite cyclic groups and derive a general formula for the corresponding unnormalized S -matrix associated with the Hopf link. We then extend this formula to arbitrary finite abelian groups. In the abelian setting, we also compute the invariant of a more complicated two-component link and obtain a general formula for links with an arbitrary number of components.

We subsequently consider the Drinfeld double of an arbitrary finite group. Using the classification of its representations, we determine the corresponding dual representations and derive formulas for the action of the universal R -matrix on different combinations of representations and their duals. These formulas lead to a general expression for the unnormalized S -matrix of the Hopf link. We illustrate the non-abelian case by explicitly computing the S -matrix associated with S_3 .

The thesis is organized as follows. Chapter 2 introduces the categorical and algebraic background required throughout the thesis, including additive, abelian, braided, and ribbon categories, as well as the graphical calculus of ribbon diagrams. Chapter 3 develops the link invariants associated with Drinfeld doubles of finite groups, beginning with the abelian case and then treating the general non-abelian case. The thesis concludes with a summary of the results and some directions for further investigation.

CHAPTER 2

Categorical and Algebraic Preliminaries

We first quickly define a category and a functor. We follow the standard categorical terminology of [5, 2].

DEFINITION 2.1. A *category* $\mathcal{C}$ consists of

- a class of objects $\text{Ob}(\mathcal{C})$;
- for every pair of objects a, b , a set of morphisms

$$\text{Hom}_{\mathcal{C}}(a, b);$$

- a composition law

$$\text{Hom}_{\mathcal{C}}(b, c) \times \text{Hom}_{\mathcal{C}}(a, b) \longrightarrow \text{Hom}_{\mathcal{C}}(a, c), \quad (g, f) \mapsto g \circ f;$$

- for each object a , an identity morphism

$$1_a \in \text{Hom}_{\mathcal{C}}(a, a),$$

such that composition is associative, and the identity morphisms act as two-sided identities.

DEFINITION 2.2. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns

- to each object $a \in \mathcal{C}$ an object $F(a) \in \mathcal{D}$;
- to each morphism $f : a \rightarrow b$ a morphism

$$F(f) : F(a) \rightarrow F(b),$$

such that

$$F(g \circ f) = F(g) \circ F(f), \quad F(1_a) = 1_{F(a)}.$$

DEFINITION 2.3 (Functorial Morphisms). Given functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a functorial morphism $\Phi : F \rightarrow G$ is a collection $\Phi_a : Fa \rightarrow Ga$ of morphisms in $\mathcal{D}$, for each object $a \in \text{Ob } \mathcal{C}$, such that for any $f \in \text{Hom}_{\mathcal{C}}(a, b)$, the following diagram commutes

$$\begin{array}{ccc} Fa & \xrightarrow{Ff} & Fb \\ \Phi_a \downarrow & & \downarrow \Phi_b \\ Ga & \xrightarrow{Gf} & Gb \end{array}$$

A functorial morphism Φ is called a *Functorial Isomorphism* if each morphism Φ_a is an isomorphism.

1. Additive and Abelian Categories

In this section, we will motivate and define an additive category, and then define the abelian category. The definitions of additive and abelian categories used in this section follow [2]. We will also give details to show that the category of linear representations of a k -algebra over a field k is abelian. This will lay the foundations for later defining a monoidal category in the next

section. Let $\mathbf{Vect}_k$ denote the category of vector spaces over k , i.e. the category whose objects are k -vector spaces and whose morphisms are k -linear maps.

In the category of vector spaces, hom-sets naturally carry the structure of a k -vector space. Moreover, given vector spaces V_1, V_2 and V_3 , the composition map

$$\mathrm{Hom}(V_2, V_3) \times \mathrm{Hom}(V_1, V_2) \longrightarrow \mathrm{Hom}(V_1, V_3)$$

is bilinear. Furthermore, the zero vector space is both an initial and a terminal object, and finite direct sums of objects exist. We formalize these properties in the following definitions.

DEFINITION 2.4. [2] A category $\mathcal{C}$ is a *k-linear additive category* if, in addition, it satisfies:

- for any objects $a, b, c \in \mathrm{Ob} \mathcal{C}$, the set $\mathrm{Hom}_{\mathcal{C}}(a, b)$ is a k -vector space, and the composition map

$$\begin{aligned} \circ : \mathrm{Hom}_{\mathcal{C}}(b, c) \times \mathrm{Hom}_{\mathcal{C}}(a, b) &\longrightarrow \mathrm{Hom}_{\mathcal{C}}(a, c) \\ (g, f) &\longmapsto g \circ f \end{aligned}$$

is k -bilinear.

- there exists an object $0 \in \mathrm{Ob} \mathcal{C}$ such that

$$\mathrm{Hom}_{\mathcal{C}}(0, a) = \mathrm{Hom}_{\mathcal{C}}(a, 0) = \{0\},$$

i.e. 0 is both an initial and a terminal object in $\mathcal{C}$.

- finite direct sums of objects in $\mathcal{C}$ exist.

In the category $\mathbf{Vect}_k$, there is a zero object and hence zero morphisms. This allows us to define the notions of kernel and cokernel, which exist for every linear map. More generally, in any category with a zero object, kernels and cokernels can be defined as follows.

DEFINITION 2.5. Given a morphism $\varphi : a \rightarrow b$, a *kernel* of φ is a morphism $i : K \rightarrow a$ with $\varphi \circ i = 0$ such that for any morphism $f : X \rightarrow a$ with $\varphi \circ f = 0$, there exists a unique morphism $\tilde{f} : X \rightarrow K$ with $f = i \circ \tilde{f}$.

$$\begin{array}{ccccc} & & X & \longrightarrow & 0 \\ & \exists! \tilde{f} \nearrow & \downarrow f & & \downarrow \\ K & \xrightarrow{i} & a & \xrightarrow{\varphi} & b \end{array}$$

Similarly, a *cokernel* of φ is a morphism $p : b \rightarrow C$ with $p \circ \varphi = 0$ such that for any morphism $g : b \rightarrow Y$ with $g \circ \varphi = 0$, there exists a unique morphism $\tilde{g} : C \rightarrow Y$ with $g = \tilde{g} \circ p$.

$$\begin{array}{ccccc} & & Y & \longleftarrow & 0 \\ & \exists! \tilde{g} \nwarrow & \uparrow g & & \uparrow \\ C & \xleftarrow{p} & b & \xleftarrow{\varphi} & a \end{array}$$

DEFINITION 2.6. A morphism $\varphi : a \rightarrow b$ is called a *monomorphism* if it is *left-cancellative*, i.e. for any object X and any morphisms $f, g : X \rightarrow a$,

$$\varphi \circ f = \varphi \circ g \implies f = g.$$

Similarly, a morphism $\varphi : a \rightarrow b$ is called an *epimorphism* if it is *right-cancellative*, i.e. for any object Y and any morphisms $f, g : b \rightarrow Y$,

$$f \circ \varphi = g \circ \varphi \implies f = g.$$

DEFINITION 2.7. An additive category $\mathcal{C}$ is called *abelian* if every morphism φ has a kernel and a cokernel, every morphism can be written as a composition $\varphi = m \circ e$, where e is an epimorphism and m is a monomorphism, and the following hold:

- if $\ker(\varphi) = 0$, then φ is a kernel of $\text{coker}(\varphi)$;
- if $\text{coker}(\varphi) = 0$, then φ is a cokernel of $\ker(\varphi)$.

DEFINITION 2.8. An object U in an abelian category $\mathcal{C}$ is called *simple* if any injection $V \hookrightarrow U$ is either 0 or an isomorphism.

DEFINITION 2.9. An *algebra* A over a field k is a k -vector space equipped with a multiplication map

$$\cdot : A \times A \longrightarrow A$$

such that $(A, +, \cdot)$ is a unital ring and

$$a \cdot (\lambda b) = (\lambda a) \cdot b = \lambda(a \cdot b)$$

for all $a, b \in A$ and $\lambda \in k$.

The algebra A is called *commutative* if

$$a \cdot b = b \cdot a$$

for all $a, b \in A$.

Equivalent definition of a k -algebra.

DEFINITION 2.10. Following [6, Definition A.3], an algebra (A, m, η) over a field k is a vector space A over k together with linear maps

$$m : A \otimes A \longrightarrow A \quad \text{and} \quad \eta : k \longrightarrow A,$$

the multiplication and the unit, such that the following two diagrams commute:

$$\begin{array}{ccccc} A \otimes (A \otimes A) & \xrightarrow{\cong} & (A \otimes A) \otimes A & \xrightarrow{m \otimes \text{id}} & A \otimes A \\ & \searrow \text{id} \otimes m & & & \downarrow m \\ & & A \otimes A & \xrightarrow{m} & A. \end{array}$$

associativity

$$\begin{array}{ccccc} k \otimes A & \xrightarrow[\cong]{\lambda \otimes a \mapsto \lambda a} & A & \xleftarrow[\cong]{a \otimes \lambda \mapsto \lambda a} & A \otimes k \\ & \searrow \eta \otimes \text{id} & \uparrow m & \swarrow \text{id} \otimes \eta & \\ & & A \otimes A & & \end{array}$$

unitality

An algebra A is called *commutative* if

$$m^{\text{op}} := m \circ \tau = m,$$

where

$$\tau : A \otimes A \longrightarrow A \otimes A, \quad a \otimes b \longmapsto b \otimes a,$$

is called the *flip map*.

DEFINITION 2.11. Let A and B be algebras over k . An *algebra homomorphism*

$$\varphi : A \longrightarrow B$$

is a k -linear map that is also a unital ring homomorphism. Thus,

$$\varphi(1_A) = 1_B$$

and

$$\varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in A$.

Similarly, coalgebras are defined by reversing the structure maps appearing in the definition of an algebra.

DEFINITION 2.12. Following [6, Definition 5.1.1], a *coalgebra* (C, Δ, ε) over a field k is a vector space C over k together with linear maps

$$\Delta : C \longrightarrow C \otimes C \quad \text{and} \quad \varepsilon : C \longrightarrow k,$$

called the *comultiplication* and the *counit*, respectively, such that the following diagrams commute:

$$\begin{array}{ccc} & C & \\ \Delta \swarrow & & \searrow \Delta \\ C \otimes C & & C \otimes C \\ \Delta \otimes \text{id} \downarrow & & \downarrow \text{id} \otimes \Delta \\ (C \otimes C) \otimes C & \xrightarrow{\cong} & C \otimes (C \otimes C). \end{array}$$

coassociativity

Let

$$\lambda_C : k \otimes C \longrightarrow C, \quad \lambda \otimes c \longmapsto \lambda c,$$

and

$$\rho_C : C \otimes k \longrightarrow C, \quad c \otimes \lambda \longmapsto \lambda c.$$

Then

$$\begin{array}{ccccc} & C & & & \\ \Delta \swarrow & & \downarrow \text{id}_C & & \searrow \Delta \\ C \otimes C & \xrightarrow{\lambda_C \circ (\varepsilon \otimes \text{id})} & C & \xleftarrow{\rho_C \circ (\text{id} \otimes \varepsilon)} & C \otimes C \end{array}$$

commutes.

counitality

A coalgebra C is called *cocommutative* if

$$\Delta^{\text{op}} := \tau \circ \Delta = \Delta.$$

DEFINITION 2.13. A finite-dimensional *linear representation* of a k -algebra A is a finite-dimensional k -vector space V , together with an algebra homomorphism

$$\rho : A \longrightarrow \text{End}_k(V).$$

Equivalently, we can define an algebra representation as a left A -module.

DEFINITION 2.14. Given a k -algebra A and representations $\rho_i : A \longrightarrow \text{End}_k(V_i)$ for $i = 1, 2$, an A -linear map $\phi : V_1 \longrightarrow V_2$ is a linear map such that for each $a \in A$, $\phi \circ \rho_1(a) = \rho_2(a) \circ \phi$.

Equivalently, A -linear maps between two left A -modules are module morphisms.

DEFINITION 2.15. Given a k -algebra A , the corresponding category of representations $\text{Rep}(A)$ is the category with objects being linear representations of A , and the morphisms A -linear maps.

REMARK 2.1. Let A be a k -algebra. Then the category $\text{Rep}(A)$ of representations of A is an abelian category.

We now verify the assertion in Remark 2.1.

1. The category $\text{Rep}(A)$ is k -linear and additive.

Consider two representations (ρ_1, V_1) and (ρ_2, V_2) . Let

$$\phi, \varphi \in \text{Hom}_{\text{Rep}(A)}(V_1, V_2), \quad \lambda \in k, \quad a \in A, \quad v \in V_1.$$

Then

$$\begin{aligned} (\lambda\phi)(\rho_1(a)v) &= \lambda\phi(\rho_1(a)v) \\ &= \lambda\rho_2(a)\phi(v) \\ &= \rho_2(a)\lambda\phi(v) \\ &= \rho_2(a)(\lambda\phi)(v). \end{aligned}$$

Also,

$$\begin{aligned} (\phi + \varphi)(\rho_1(a)v) &= \phi(\rho_1(a)v) + \varphi(\rho_1(a)v) \\ &= \rho_2(a)\phi(v) + \rho_2(a)\varphi(v) \\ &= \rho_2(a)(\phi + \varphi)(v). \end{aligned}$$

Therefore,

$$\text{Hom}_{\text{Rep}(A)}(V_1, V_2)$$

is a k -vector space. The composition of A -linear maps is k -bilinear.

The zero vector space, equipped with the unique A -action

$$\rho_0 : A \longrightarrow \text{End}_k(0) = \{0\},$$

is a zero object in $\text{Rep}(A)$.

Now let (ρ_i, V_i) , for $i = 1, \dots, n$, be a finite collection of representations. Define an action of A on

$$V_1 \oplus \dots \oplus V_n$$

by

$$\rho(a)(v_1, \dots, v_n) = (\rho_1(a)v_1, \dots, \rho_n(a)v_n).$$

This defines a representation because

$$\begin{aligned} \rho(ab)(v_1, \dots, v_n) &= (\rho_1(ab)v_1, \dots, \rho_n(ab)v_n) \\ &= (\rho_1(a)\rho_1(b)v_1, \dots, \rho_n(a)\rho_n(b)v_n) \\ &= \rho(a)\rho(b)(v_1, \dots, v_n), \end{aligned}$$

and

$$\rho(1_A)(v_1, \dots, v_n) = (v_1, \dots, v_n).$$

Thus finite direct sums of representations are again representations of A . Hence $\text{Rep}(A)$ is a k -linear additive category.

2. Kernels exist in $\text{Rep}(A)$.

Let

$$\varphi : V_1 \longrightarrow V_2$$

be an A -linear map. If $v \in \ker(\varphi)$, then

$$\varphi(\rho_1(a)v) = \rho_2(a)\varphi(v) = 0.$$

Therefore,

$$\ker(\varphi) = \{v \in V_1 \mid \varphi(v) = 0\}$$

is stable under the action of A via ρ_1 , and hence defines an object, say $(\tilde{\rho}_1, \ker(\varphi))$. The inclusion

$$i : \ker(\varphi) \hookrightarrow V_1$$

is A -linear.

Suppose that $f : X \longrightarrow V_1$ is an A -linear map satisfying

$$\varphi \circ f = 0.$$

Then $f(X) \subseteq \ker(\varphi)$. Define

$$\tilde{f} : X \longrightarrow \ker(\varphi), \quad \tilde{f}(x) = f(x).$$

This map is A -linear and satisfies

$$f = i \circ \tilde{f}.$$

It is unique because i is injective. Therefore, i is the kernel of φ in $\text{Rep}(A)$.

3. Cokernels exist in $\text{Rep}(A)$.

If $w = \varphi(v) \in \text{im}(\varphi)$, then

$$\rho_2(a)w = \rho_2(a)\varphi(v) = \varphi(\rho_1(a)v).$$

Therefore, $\text{im}(\varphi)$ is stable under the action of A , so it is an A -submodule of V_2 .

Define an action of A on

$$V_2 / \text{im}(\varphi)$$

by

$$a \cdot (v + \text{im}(\varphi)) = \rho_2(a)v + \text{im}(\varphi).$$

This action is well-defined. Indeed, if

$$v + \text{im}(\varphi) = v' + \text{im}(\varphi),$$

then

$$v - v' \in \text{im}(\varphi).$$

Since $\text{im}(\varphi)$ is stable under the action of A , we have

$$\rho_2(a)(v - v') \in \text{im}(\varphi).$$

Hence

$$\rho_2(a)v + \text{im}(\varphi) = \rho_2(a)v' + \text{im}(\varphi).$$

The canonical projection

$$p : V_2 \longrightarrow V_2 / \text{im}(\varphi), \quad p(v) = v + \text{im}(\varphi),$$

is A -linear because

$$p(\rho_2(a)v) = \rho_2(a)v + \text{im}(\varphi) = a \cdot p(v).$$

Suppose that $g : V_2 \longrightarrow Y$ is an A -linear map satisfying

$$g \circ \varphi = 0.$$

Then g vanishes on $\text{im}(\varphi)$. Define

$$\tilde{g} : V_2 / \text{im}(\varphi) \longrightarrow Y$$

by

$$\tilde{g}(v + \text{im}(\varphi)) = g(v).$$

This map is well-defined and A -linear, and it satisfies

$$g = \tilde{g} \circ p.$$

It is unique because p is surjective. Therefore, p is the cokernel of φ in $\text{Rep}(A)$.

4. Every morphism factors as an epimorphism followed by a monomorphism.

The morphism φ factors as

$$V_1 \xrightarrow{q} V_1 / \ker(\varphi) \xrightarrow{\bar{\varphi}} \text{im}(\varphi) \xrightarrow{j} V_2,$$

where q is the canonical projection, j is the inclusion, and

$$\bar{\varphi}(v + \ker(\varphi)) = \varphi(v).$$

The map $\bar{\varphi}$ is an isomorphism of representations. Therefore,

$$\varphi = j \circ \bar{\varphi} \circ q,$$

where q is an epimorphism and

$$j \circ \bar{\varphi} : V_1 / \ker(\varphi) \longrightarrow V_2$$

is a monomorphism.

5. Monomorphisms and epimorphisms are normal.

If

$$\ker(\varphi) = 0,$$

then φ identifies V_1 with $\text{im}(\varphi)$. Since

$$\text{im}(\varphi) = \ker(V_2 \longrightarrow V_2 / \text{im}(\varphi)),$$

the morphism φ is, up to the canonical identification, the kernel of its cokernel.

Similarly, if

$$\text{coker}(\varphi) = 0,$$

then

$$\text{im}(\varphi) = V_2,$$

so φ is surjective. Since

$$V_1 / \ker(\varphi) \cong V_2,$$

the morphism φ is, up to the canonical identification, the cokernel of its kernel.

Hence $\text{Rep}(A)$ is an abelian category.

2. Braided and Ribbon Categories

In this section, we recall the definitions of a monoidal category and a braided tensor category, following [2].

DEFINITION 2.16. Given categories $\mathcal{C}$ and $\mathcal{D}$, the *product category* $\mathcal{C} \times \mathcal{D}$ is a category with objects of the form $c \times d$, where $c \in \text{Ob } \mathcal{C}$ and $d \in \text{Ob } \mathcal{D}$ and for morphisms $f : c \rightarrow c'$ in $\mathcal{C}$ and $g : d \rightarrow d'$ in $\mathcal{D}$, a morphism $f \times g : c \times d \rightarrow c' \times d'$.

It is easy to check that the product category is a category with composition in the obvious way and for each element $c \times d$, the identity morphism $1_c \times 1_d$, where 1_c is the identity morphism of c in $\mathcal{C}$ and 1_d is the identity morphism of d in $\mathcal{D}$. A functor from a product category is called a *bifunctor*.

DEFINITION 2.17. A monoidal category is a category $\mathcal{C}$, with the following additional data:

- a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$.
- a functorial isomorphism,

$$\alpha_{a,b,c} : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c)$$

between the functors

$$\otimes \circ (\otimes \times \text{id}_{\mathcal{C}}), \otimes \circ (\text{id}_{\mathcal{C}} \times \otimes) : \mathcal{C} \times \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$$

- an element $1 \in \text{Ob } \mathcal{C}$ called *unit object* and functorial isomorphisms

$$\lambda_a : 1 \otimes a \rightarrow a$$

and

$$\rho_a : a \otimes 1 \rightarrow a$$

satisfying the coherence axiom:

- given an object $a_1 \otimes a_2 \otimes \cdots \otimes a_n$ and for any two expressions obtained from $a_1 \otimes a_2 \otimes \cdots \otimes a_n$ by putting 1's and brackets in grammatically correct ways, there is a unique isomorphism as a composition of α 's, λ 's and ρ 's and their inverses.

The equivalent formulation in terms of the pentagon and triangle axioms can be found in [5, Chapter VII].

We have seen that, for a k -algebra A , the category $\text{Rep}(A)$ is an abelian category. We now want to turn $\text{Rep}(A)$ into a monoidal category.

Let (ρ_V, V) and (ρ_W, W) be two representations of A . The natural candidate for their tensor product is the vector space $V \otimes W$. One might first try to define an action of A on $V \otimes W$ by

$$a \cdot (v \otimes w) = (a \cdot v) \otimes (a \cdot w) = \rho_V(a)v \otimes \rho_W(a)w. \quad (1)$$

However, this does not, in general, define a representation of A . Indeed, for $\lambda \in k$, we obtain

$$\begin{aligned} (\lambda a) \cdot (v \otimes w) &= (\rho_V(\lambda a)v) \otimes (\rho_W(\lambda a)w) \\ &= \lambda \rho_V(a)v \otimes \lambda \rho_W(a)w \\ &= \lambda^2 (\rho_V(a)v \otimes \rho_W(a)w), \end{aligned}$$

whereas linearity would require

$$(\lambda a) \cdot (v \otimes w) = \lambda (a \cdot (v \otimes w)).$$

Thus the action in (1) is not linear in a .

Therefore, the algebra structure of A alone does not provide a canonical action on $V \otimes W$. We need an additional map

$$\Delta : A \longrightarrow A \otimes A$$

which separates the action of an element of A into actions on the two tensor factors. This motivates the definition of a bialgebra.

For the definitions and basic properties of bialgebras, Hopf algebras, and quasitriangular Hopf algebras, we follow [6].

DEFINITION 2.18. A *bialgebra* over a field k is a quintuple $(B, m, \eta, \Delta, \varepsilon)$ such that (B, m, η) is an algebra over k , (B, Δ, ε) is a coalgebra over k , and the maps

$$\Delta : B \longrightarrow B \otimes B \quad \text{and} \quad \varepsilon : B \longrightarrow k$$

are algebra homomorphisms.

Notation: We use Sweedler notation

$$\Delta(b) = \sum b_{(1)} \otimes b_{(2)},$$

where the summation symbol denotes the decomposition of $\Delta(b)$ into a sum of simple tensors.

Once we have a bialgebra structure on B , we can define a monoidal structure on $\text{Rep}(B)$ as follows. Given two representations

$$\rho_1 : B \longrightarrow \text{End}_k(V_1), \quad \rho_2 : B \longrightarrow \text{End}_k(V_2),$$

define

$$\rho_1 \otimes \rho_2 : B \longrightarrow \text{End}_k(V_1 \otimes V_2)$$

by

$$(\rho_1 \otimes \rho_2)(a)(v_1 \otimes v_2) = \sum_{(a)} \rho_1(a_{(1)})v_1 \otimes \rho_2(a_{(2)})v_2.$$

We check that this defines a representation. Since Δ is an algebra homomorphism,

$$\Delta(ab) = \Delta(a)\Delta(b) = \sum_{(a),(b)} a_{(1)}b_{(1)} \otimes a_{(2)}b_{(2)}.$$

Therefore,

$$\begin{aligned} (\rho_1 \otimes \rho_2)(ab)(v_1 \otimes v_2) &= \sum_{(a),(b)} \rho_1(a_{(1)}b_{(1)})v_1 \otimes \rho_2(a_{(2)}b_{(2)})v_2 \\ &= \sum_{(a),(b)} \rho_1(a_{(1)})\rho_1(b_{(1)})v_1 \otimes \rho_2(a_{(2)})\rho_2(b_{(2)})v_2 \\ &= (\rho_1 \otimes \rho_2)(a) ((\rho_1 \otimes \rho_2)(b)(v_1 \otimes v_2)). \end{aligned}$$

Also, since $\Delta(1_B) = 1_B \otimes 1_B$, we have

$$(\rho_1 \otimes \rho_2)(1_B) = \text{id}_{V_1 \otimes V_2}.$$

Hence $V_1 \otimes V_2$ is a representation of B .

If $f : V_1 \longrightarrow V'_1$, $g : V_2 \longrightarrow V'_2$ are B -linear maps, then

$$f \otimes g : V_1 \otimes V_2 \longrightarrow V'_1 \otimes V'_2$$

is also B -linear. Thus the tensor product defines a bifunctor

$$\otimes : \text{Rep}(B) \times \text{Rep}(B) \longrightarrow \text{Rep}(B).$$

The usual associativity isomorphism

$$\alpha_{V_1, V_2, V_3} : (V_1 \otimes V_2) \otimes V_3 \longrightarrow V_1 \otimes (V_2 \otimes V_3)$$

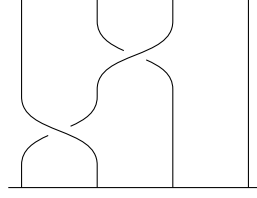


FIGURE 2.1. Example of a braid on 4 strands.

is B -linear because

$$(\Delta \otimes \text{id}) \circ \Delta = (\text{id} \otimes \Delta) \circ \Delta.$$

Finally, the unit object is k , equipped with the representation

$$\rho_k : B \longrightarrow \text{End}_k(k), \quad \rho_k(a)(\lambda) = \varepsilon(a)\lambda.$$

The counit identities imply that the usual isomorphisms

$$k \otimes V \cong V, \quad V \otimes k \cong V$$

are B -linear. Hence $\text{Rep}(B)$ is a monoidal category.

We now define the braided tensor structure.

DEFINITION 2.19. An n -strand braid is an isotopy class of a union of n pairwise non-intersecting smooth curves, called *strands*, in $\mathbb{R}^3$, with endpoints

$$\{1, \dots, n\} \times \{0\} \times \{0, 1\},$$

such that, along each strand, the third coordinate z is strictly increasing from 0 to 1. Thus, the strands are considered to go upward.

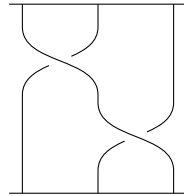
An example of a braid on 4 strands is shown in the following diagram

Braids in n strands form a group, where the group operation is defined by stacking braids upon each other.

EXAMPLE 2.1. Let $n = 3$, consider the braids b and b' as shown in Figure 2.2.

FIGURE 2.2. The braids b and b' in B_3 .

The product bb' is obtained by concatenation, shown in Figure 2.3.

FIGURE 2.3. The product bb' in B_3 .

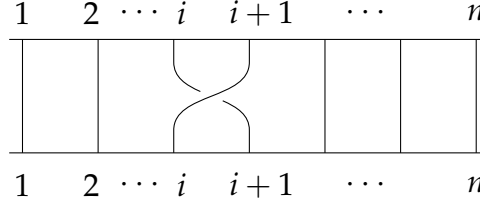
The identity braid is shown in Figure 2.4.

The following theorem due to Emil Artin [1] gives the generators b_i of B_n and braid relations.

THEOREM 2.1. The braid group B_n is generated by braids b_i as shown in Figure 2.5 for $i = 1, \dots, n - 1$ and furthermore satisfies the following relations:

FIGURE 2.4. The identity braid in B_3 .

- $b_i b_j = b_j b_i$ whenever $|i - j| > 1$
- $b_i b_{i+1} b_i = b_{i+1} b_i b_{i+1}$, $i = 1, \dots, n - 2$

FIGURE 2.5. The generator braid b_i in B_n .

Let $\mathcal{C}$ be a monoidal category, and let

$$\sigma_{a,b}: a \otimes b \longrightarrow b \otimes a, \quad a, b \in \text{Ob } \mathcal{C},$$

be a family of functorial isomorphisms.

Given objects

$$a_1, \dots, a_n \in \text{Ob } \mathcal{C},$$

consider all expressions obtained from

$$a_{i_1} \otimes \dots \otimes a_{i_n}$$

by inserting copies of the unit object $\mathbf{1}$ and brackets, where $(i_1, \dots, i_n)$ is a permutation of $(1, \dots, n)$.

Let X_1 and X_2 be two such expressions. To an isomorphism

$$\varphi: X_1 \longrightarrow X_2$$

obtained by composing α 's, λ 's, ρ 's, σ 's, and their inverses, we associate an element of B_n as follows. To a morphism of the form $\sigma_{a_{i_k}, a_{i_{k+1}}}$, we assign the braid generator b_k , whereas to α , λ , and ρ , we assign the identity element $1 \in B_n$. Inverses are assigned the corresponding inverse braid elements, and compositions are assigned the corresponding products.

We follow [2, Chapter 1] for braided tensor categories and [2, Chapter 2] for rigid and ribbon categories.

DEFINITION 2.20. Following [2, Definition 1.2.3], a *braided tensor category* $\mathcal{C}$ is a category with $\mathcal{C}, \otimes, \mathbf{1}, \alpha, \lambda, \rho, \sigma$ such that every isomorphism between two expressions described above, obtained by composing the structural isomorphisms and their inverses, depends only on its associated element of B_n . Equivalently, two such compositions associated with the same braid define the same isomorphism.

DEFINITION 2.21. Following [6, Definition 5.2.1], a *Hopf algebra* $(H, m, \eta, \Delta, \varepsilon, S)$, where $(H, m, \eta, \Delta, \varepsilon)$ is a bialgebra and $S: H \longrightarrow H$ is a linear map called *antipode* that satisfies

$$m \circ (S \otimes id) \circ \Delta = m \circ (id \otimes S) \circ \Delta = \eta \circ \varepsilon.$$

To define a braided tensor structure on $\text{Rep}(B)$, a bialgebra structure alone is not enough. We therefore introduce the notion of a quasi-triangular bialgebra. A quasi-triangular bialgebra is a bialgebra B equipped with an invertible element

$$R \in B \otimes B,$$

called the *universal R -matrix*. The universal R -matrix is required to satisfy certain compatibility conditions with the comultiplication, and these conditions ensure that, for any two representations V_1 and V_2 , R defines a braiding

$$\sigma_{V_1, V_2} : V_1 \otimes V_2 \longrightarrow V_2 \otimes V_1.$$

This motivates the following definition.

DEFINITION 2.22 ([6, Definition 6.3.1]). A *quasi-triangular bialgebra* is a pair (B, R) of a bialgebra B and an invertible element $R \in B \otimes B$ satisfying the following properties:

- $\Delta^{op}(b) = R \cdot \Delta(b) \cdot R^{-1}$
- $(\Delta \otimes id)(R) = R_{13} \cdot R_{23}$
- $(id \otimes \Delta)(R) = R_{13} \cdot R_{12}$

where $R_{12} = \sum R_{(1)} \otimes R_{(2)} \otimes 1$, $R_{13} = \sum R_{(1)} \otimes 1 \otimes R_{(2)}$ and $R_{23} = \sum 1 \otimes R_{(1)} \otimes R_{(2)} \in B \otimes B \otimes B$

The representation of the universal R -matrix, together with the flip map, gives the braiding

$$\sigma_{V_1, V_2} : V_1 \otimes V_2 \longrightarrow V_2 \otimes V_1,$$

defined by

$$\sigma_{V_1, V_2} = \tau \circ (\rho_1 \otimes \rho_2)(R),$$

where

$$\tau(v_1 \otimes v_2) = v_2 \otimes v_1.$$

For instance, if we consider the braided tensor category $\mathbf{Vect}_k$, with

$$\sigma_{V_1, V_2}(v_1 \otimes v_2) = v_2 \otimes v_1,$$

then

$$\sigma_{V_2, V_1} \circ \sigma_{V_1, V_2} = \text{id}_{V_1 \otimes V_2}.$$

In terms of the braid group action, the braid b_1^2 is not the identity braid in B_2 , but its image under the corresponding representation is the identity morphism. Thus, in $\mathbf{Vect}_k$, the double braiding is trivial, and the braiding is symmetric.

In a general braided tensor category arising from a quasi-triangular bialgebra, however, we need not have

$$\sigma_{V_2, V_1} \circ \sigma_{V_1, V_2} = \text{id}_{V_1 \otimes V_2}.$$

Thus the braid b_1^2 may act nontrivially. To construct examples of braided tensor categories, we consider the Drinfeld double/Quantum double of finite groups

DEFINITION 2.23 ([6, Example 6.3.7]). Let G be a finite group. The Drinfeld double $D(G)$ is the vector space

$$\text{Map}(G, k) \otimes_k k[G]$$

with the quasitriangular Hopf algebra structure given by

- $(\delta_u \otimes g) \cdot (\delta_v \otimes h) = \delta_u(gv g^{-1}) \delta_u \otimes gh,$
- $\Delta(\delta_u \otimes g) = \sum_{xy=u} (\delta_y \otimes g) \otimes (\delta_x \otimes g),$
- $S(\delta_u \otimes g) = \delta_{g^{-1}u^{-1}g} \otimes g^{-1},$
- $1 = 1 \otimes e = \sum_{g \in G} \delta_g \otimes e,$

- $\epsilon(\delta_u \otimes g) = \delta_u(e)$,
- $R = \sum_{g \in G} (1 \otimes g) \otimes (\delta_g \otimes e)$.

Since $D(G)$ is also equipped with the antipode S , it is a quasitriangular Hopf algebra.

The Drinfeld double of a finite-dimensional semisimple Hopf algebra over a field of characteristic zero is a ribbon Hopf algebra [4]. Since the group algebra $k[G]$ of a finite group G is finite-dimensional and semisimple when $\text{char}(k) = 0$, its Drinfeld double $D(G)$ is a ribbon Hopf algebra. Consequently, the category of finite-dimensional representations

$$\text{Rep}(D(G))$$

is a ribbon category.

DEFINITION 2.24. Following [2, Definition 2.1.1], let $(\mathcal{C}, \otimes, 1, \alpha, \lambda, \rho)$ be a monoidal category and let V be an object in $\mathcal{C}$. A *right dual* of V is an object V^* , together with two morphisms

$$e_V : V^* \otimes V \longrightarrow 1, \quad (2)$$

and

$$i_V : 1 \longrightarrow V \otimes V^*, \quad (3)$$

such that the composition

$$V \xrightarrow{\lambda_V^{-1}} 1 \otimes V \xrightarrow{i_V \otimes \text{id}_V} (V \otimes V^*) \otimes V \xrightarrow{\alpha_{V, V^*, V}} V \otimes (V^* \otimes V) \xrightarrow{\text{id}_V \otimes e_V} V \otimes 1 \xrightarrow{\rho_V} V \quad (4)$$

is equal to id_V , and the composition

$$V^* \xrightarrow{\rho_{V^*}^{-1}} V^* \otimes 1 \xrightarrow{\text{id}_{V^*} \otimes i_V} V^* \otimes (V \otimes V^*) \xrightarrow{\alpha_{V^*, V, V^*}^{-1}} (V^* \otimes V) \otimes V^* \xrightarrow{e_V \otimes \text{id}_{V^*}} 1 \otimes V^* \xrightarrow{\lambda_{V^*}} V^* \quad (5)$$

is equal to id_{V^*} .

DEFINITION 2.25. Following [2, Definition 2.1.1], $(\mathcal{C}, \otimes, 1, \alpha, \lambda, \rho)$ be a monoidal category and let V be an object in $\mathcal{C}$. A *left dual* of V is an object *V , together with two morphisms

$$e'_V : V \otimes {}^*V \longrightarrow 1, \quad (6)$$

and

$$i'_V : 1 \longrightarrow {}^*V \otimes V, \quad (7)$$

such that the composition

$${}^*V \xrightarrow{\lambda_{{}^*V}^{-1}} 1 \otimes {}^*V \xrightarrow{i'_V \otimes \text{id}_{{}^*V}} ({}^*V \otimes V) \otimes {}^*V \xrightarrow{\alpha_{{}^*V, V, {}^*V}} {}^*V \otimes (V \otimes {}^*V) \xrightarrow{\text{id}_{{}^*V} \otimes e'_V} {}^*V \otimes 1 \xrightarrow{\rho_{{}^*V}} {}^*V \quad (8)$$

is equal to $\text{id}_{{}^*V}$, and the composition

$$V \xrightarrow{\rho_V^{-1}} V \otimes 1 \xrightarrow{\text{id}_V \otimes i'_V} V \otimes ({}^*V \otimes V) \xrightarrow{\alpha_{V, {}^*V, V}^{-1}} (V \otimes {}^*V) \otimes V \xrightarrow{e'_V \otimes \text{id}_V} 1 \otimes V \xrightarrow{\lambda_V} V \quad (9)$$

is equal to id_V .

DEFINITION 2.26. A monoidal category is *rigid* if each object has a left and a right dual.

For example, the category $\mathbf{Vect}_k^{\text{fd}}$ of finite-dimensional vector spaces over k is rigid.

DEFINITION 2.27. A *ribbon category* is a rigid braided tensor category equipped with a functorial isomorphism

$$\delta_V : V \longrightarrow V^{**}$$

satisfying the following conditions:

-
-
-

$$\delta_{V \otimes W} = \delta_V \otimes \delta_W;$$

$$\delta_1 = \text{id}_1;$$

$$\delta_{V^*} = ((\delta_V)^*)^{-1}.$$

Here we use the canonical identifications

$$(V \otimes W)^{**} \cong V^{**} \otimes W^{**} \quad \text{and} \quad 1^{**} \cong 1.$$

For a morphism $f \in \text{Hom}_{\mathcal{C}}(V, W)$, the dual morphism $f^* \in \text{Hom}_{\mathcal{C}}(W^*, V^*)$ is defined by the composition

$$\begin{aligned} f^* : W^* &\xrightarrow{\rho_{W^*}^{-1}} W^* \otimes 1 \\ &\xrightarrow{\text{id}_{W^*} \otimes i_V} W^* \otimes (V \otimes V^*) \\ &\xrightarrow{\alpha_{W^*, V, V^*}^{-1}} (W^* \otimes V) \otimes V^* \\ &\xrightarrow{(\text{id}_{W^*} \otimes f) \otimes \text{id}_{V^*}} (W^* \otimes W) \otimes V^* \\ &\xrightarrow{e_W \otimes \text{id}_{V^*}} 1 \otimes V^* \\ &\xrightarrow{\lambda_{V^*}} V^*. \end{aligned} \tag{10}$$

We assume that $\mathcal{C}$ is an abelian k -linear category, the tensor product is bilinear, the unit object 1 is simple, and

$$\text{End}_{\mathcal{C}}(1) = k.$$

Let $\mathcal{C}$ be a rigid braided tensor category. There is a canonical functorial isomorphism

$$\psi_V : V^{**} \longrightarrow V$$

defined by the composition

$$\begin{aligned} V^{**} &\xrightarrow{\lambda_{V^{**}}^{-1}} 1 \otimes V^{**} \xrightarrow{i_V \otimes \text{id}_{V^{**}}} (V \otimes V^*) \otimes V^{**} \xrightarrow{\alpha_{V, V^*, V^{**}}} V \otimes (V^* \otimes V^{**}) \\ &\xrightarrow{\text{id}_V \otimes \sigma_{V^{**}, V^*}^{-1}} V \otimes (V^{**} \otimes V^*) \xrightarrow{\text{id}_V \otimes e_{V^*}} V \otimes 1 \xrightarrow{\rho_V} V. \end{aligned}$$

The maps ψ_V satisfy

$$\psi_{V \otimes W} = \sigma_{W, V} \circ \sigma_{V, W} \circ (\psi_V \otimes \psi_W),$$

where we use the canonical identification $(V \otimes W)^{**} \cong V^{**} \otimes W^{**}$, and $\psi_1 = \text{id}_1$.

In general, $\psi_{V \otimes W} \neq \psi_V \otimes \psi_W$. If the braiding is symmetric, then $\sigma_{W, V} \circ \sigma_{V, W} = \text{id}_{V \otimes W}$, and hence

$$\psi_{V \otimes W} = \psi_V \otimes \psi_W.$$

Now suppose that $\mathcal{C}$ is a ribbon category with the functorial isomorphisms

$$\delta_V : V \longrightarrow V^{**}.$$

We define

$$\theta_V := \psi_V \circ \delta_V : V \longrightarrow V.$$

The family of functorial isomorphisms

$$\theta_V : V \longrightarrow V, \quad V \in \text{Ob}(\mathcal{C}),$$

is called the *balancing isomorphism*, or the *twist*.

The twist satisfies the following properties:

$$\theta_{V \otimes W} = \sigma_{W,V} \circ \sigma_{V,W} \circ (\theta_V \otimes \theta_W),$$

$$\theta_1 = \text{id}_1,$$

and

$$\theta_{V^*} = (\theta_V)^*.$$

The first identity is called the *balancing axiom*. Thus, the twist on a tensor product is not in general equal to $\theta_V \otimes \theta_W$; the additional factor

$$\sigma_{W,V} \circ \sigma_{V,W}$$

takes into account the double braiding.

Conversely, if a rigid braided tensor category is equipped with a family of functorial isomorphisms

$$\theta_V : V \longrightarrow V$$

satisfying

$$\theta_{V \otimes W} = \sigma_{W,V} \circ \sigma_{V,W} \circ (\theta_V \otimes \theta_W),$$

$$\theta_1 = \text{id}_1,$$

and

$$\theta_{V^*} = (\theta_V)^*,$$

then the twists uniquely determine the functorial isomorphisms

$$\delta_V : V \longrightarrow V^{**},$$

and hence determine a ribbon structure on $\mathcal{C}$.

In particular, if $\mathcal{C}$ is symmetric, then

$$\sigma_{W,V} \circ \sigma_{V,W} = \text{id}_{V \otimes W},$$

so the balancing axiom reduces to

$$\theta_{V \otimes W} = \theta_V \otimes \theta_W.$$

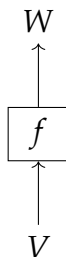
3. Graphical Calculus for Ribbon Categories

Let $\mathcal{C}$ be a ribbon category. In this section, we briefly introduce the graphical calculus for morphisms that will be used throughout the rest of this thesis. We will define only the tools we will use in the coming sections and skip details. For a detailed treatment of the graphical calculus of ribbon categories, see [2, Chapter 2]. We will use the symbol $\doteq$ to mean that two diagrams represent the same morphism.

A morphism

$$f : V \longrightarrow W$$

is represented as follows:



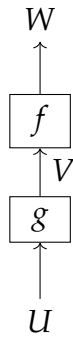
where the diagram is read from bottom to top. Similarly an identity morphism $id_V : V \longrightarrow V$ is represented as follows:



Composition of morphisms is represented by vertical concatenation. Thus, the diagram corresponding to

$$f \circ g : U \longrightarrow W$$

is represented as follows:



where $f : V \longrightarrow W$ and $g : U \longrightarrow V$.

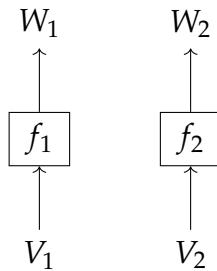
The tensor product of morphisms is represented as follows. If

$$f_1 : V_1 \longrightarrow W_1, \quad f_2 : V_2 \longrightarrow W_2,$$

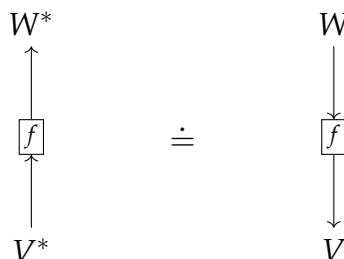
then

$$f_1 \otimes f_2 : V_1 \otimes V_2 \longrightarrow W_1 \otimes W_2$$

is represented by placing the diagram of f_1 to the left of the diagram of f_2 .



A morphism $f : V^* \longrightarrow W^*$ will be represented by reversing arrows as follows:



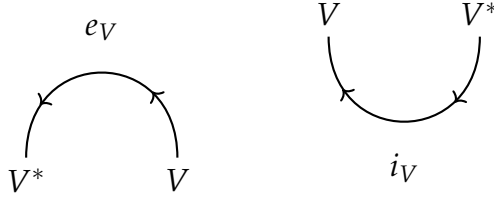
For a right dual V^* , the evaluation morphism

$$e_V : V^* \otimes V \longrightarrow 1$$

is represented by a cap, while the coevaluation morphism

$$i_V : 1 \longrightarrow V \otimes V^*$$

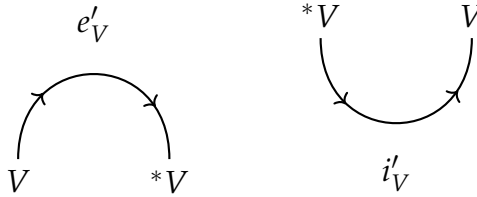
is represented by a cup.



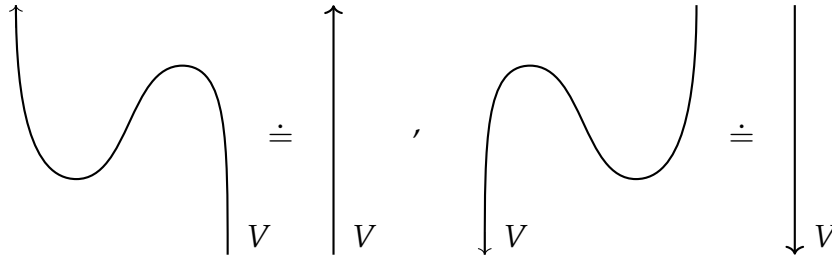
Similarly, for a left dual *V , the evaluation and coevaluation morphisms

$$e'_V : V \otimes {}^*V \longrightarrow 1, \quad i'_V : 1 \longrightarrow {}^*V \otimes V$$

are represented by the oppositely oriented cap and cup.



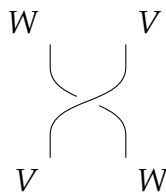
The rigidity axioms can now be interpreted graphically. They state that a strand which is bent using an evaluation and a coevaluation morphism can be straightened without changing the corresponding morphism. These identities are also called the *snake identities*.



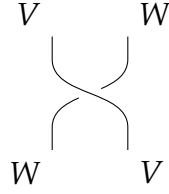
Finally, the braiding

$$\sigma_{V,W} : V \otimes W \longrightarrow W \otimes V$$

is represented by a crossing. We use the same braid notation introduced earlier.



The inverse braiding is represented by the opposite crossing.



These diagrams provide a convenient way of representing and manipulating morphisms in a ribbon category.

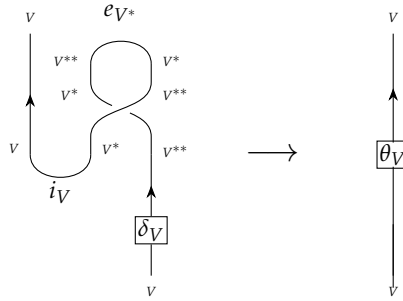
We make the following important remark.

REMARK 2.2. In a ribbon category, one can canonically choose ${}^*V = V^*$ with

$$e'_V = e_{V^*} \circ (\delta_V \otimes id_{V^*})$$

$$i'_V = (id_{V^*} \otimes \delta_V^{-1}) \circ i_{V^*}$$

3.1. Ribbons, Twists, and Framed Tangles. The morphism θ_V is represented as follows:



In tangle diagrams, we usually ignore δ_V . Therefore, the above diagram is modified to the following diagram:

w

We now introduce the notions of tangles and ribbon tangles.

DEFINITION 2.28. A *tangle* is an isotopy class of a collection of non-intersecting smooth curves in $\mathbb{R}^2 \times [0, 1]$, whose endpoints are allowed to lie only on the boundary planes $\mathbb{R}^2 \times \{0\}$ and $\mathbb{R}^2 \times \{1\}$. If all the curves are closed, the tangle is called a *link*. In particular, a link consisting of a single closed curve is called a *knot*.

The diagrams representing θ_V and id_V are isotopic as unframed tangles, although the corresponding morphisms are generally different. To retain the twisting information, we therefore consider framed, or ribbon, tangles.

DEFINITION 2.29. A *ribbon tangle*, or *framed tangle*, of n strands is an isotopy class of a union of n non-intersecting ribbons in $\mathbb{R}^2 \times [0, 1]$, such that the bases of the ribbons lie on $\mathbb{R} \times \{0\} \times \{0, 1\}$, and near these lines the ribbons have their face side turned upward.

A ribbon tangle consisting entirely of annuli is called a *ribbon link*, or *framed link*. A ribbon link consisting of a single annulus is called a *framed knot*.

A labeled and oriented framed tangle determines a morphism in the corresponding ribbon category. Informally, the Reshetikhin–Turaev theorem states that isotopic labeled and oriented framed tangles determine the same morphism [7].

CHAPTER 3

Link Invariants from Drinfeld Doubles of Finite Groups

In this chapter, we consider some cases of finite groups and their corresponding Drinfeld doubles $D(G)$. We use the Hopf link with an orientation as our main example. We color each component of the link by a representation of the Drinfeld double. This allows us to represent the link by morphisms in the corresponding representation category $\text{Rep}(D(G))$, and hence to compute the associated scalar for each pair of irreducible representations of $D(G)$. Collecting these scalars gives a matrix. Its entries are the Reshetikhin–Turaev invariants [7] of the correspondingly colored Hopf links.

From now on, we work over $k = \mathbb{C}$ and consider the category $\text{Rep}(D(G))$.

1. The Abelian Case: Finite Cyclic Groups

Consider the finite abelian group $G = \mathbb{Z}/n\mathbb{Z} \cong \{[0], [1], \dots, [n-1]\}$. The irreducible representations of $D(G)$ are given by (g, χ) with $g \in G$ and $\chi : G \rightarrow \mathbb{C}^*$ such that $\chi(a+b) = \chi(a)\chi(b)$ [3]. Furthermore,

$$\chi([0]) = \chi([0] + [0]) = \chi([0])\chi([0]) = \chi([0])^2,$$

therefore $\chi([0]) = 1$.

Also,

$$1 = \chi([n]) = \chi([1] + [1] + \dots + [1]) = \chi([1])^n,$$

therefore $\chi([1]) = e^{2\alpha\pi i/n}$ for some integer $0 \leq \alpha \leq n-1$.

Given a one-dimensional vector space $V = \text{span}\{e\}$, the representation determined by the pair (g, χ) is given by

$$\begin{aligned} \pi_{g,\chi} : D(G) &\longrightarrow \text{End}(V), \\ \delta_h \otimes k &\longmapsto \delta_{h,g} \chi(k). \end{aligned}$$

We consider the R -matrix

$$R = \sum_{g \in G} (1 \otimes g) \otimes (\delta_g \otimes e) = \sum_{g=[0]}^{[n-1]} (1 \otimes g) \otimes (\delta_g \otimes [0]),$$

where 1 is the constant map $G \rightarrow \mathbb{C}$ with value 1 and $[0]$ is the additive identity of the group G . We fix two representations (g_1, χ_1) and (g_2, χ_2) , where χ_1 is defined by $\chi_1(1) = e^{2\alpha_1\pi i/n}$ and χ_2 is defined by $\chi_2(1) = e^{2\alpha_2\pi i/n}$. The representation of R in $V_1 \otimes V_2$ is given by

$$\begin{aligned} R_{1,2} &= \sum_{g=[0]}^{[n-1]} 1 \cdot \chi_1(g) \otimes \delta_{g,g_2} \cdot \chi_2([0]) \\ &= \sum_{g=[0]}^{[n-1]} e^{2\pi g\alpha_1 i/n} \otimes \delta_{g,g_2} \cdot 1 \\ &= e^{2\pi g_2\alpha_1 i/n} \otimes 1 \\ &= \chi_1(g_2) \otimes 1 \end{aligned}$$

Similarly, the representation of R in $V_2 \otimes V_1$ is given by

$$R_{2,1} = \chi_2(g_1) \otimes 1$$

Let V be a finite-dimensional representation of $D(G)$. Let $\{v_i\}$ be a basis of V , and let $\{v^i\}$ be the corresponding dual basis of V^* . The standard right and left evaluation and coevaluation maps are given as follows (see [6, Section 5.2]):

$$e_V: V^* \otimes V \longrightarrow \mathbb{C}, \quad e_V(f \otimes v) = f(v),$$

$$i_V: \mathbb{C} \longrightarrow V \otimes V^*, \quad i_V(1) = \sum_i v_i \otimes v^i,$$

$$e'_V: V \otimes V^* \longrightarrow \mathbb{C}, \quad e'_V(v \otimes f) = f(v),$$

$$i'_V: \mathbb{C} \longrightarrow V^* \otimes V, \quad i'_V(1) = \sum_i v^i \otimes v_i.$$

We use the standard pivotal structure on $\text{Rep}(D(G))$ corresponding to the pivot 1, given by

$$\delta_V(v)(f) = f(v), \quad v \in V, \quad f \in V^*;$$

see [6, Proposition 6.4.12].

Now we consider the Hopf link as shown in Figure 3.1

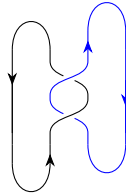


FIGURE 3.1. Hopf link with an orientation.

This link is represented as a morphism, and the corresponding values of the morphism are computed as shown in the following diagram. Note that the braiding morphism $\sigma_{1,2} = \tau \circ R_{1,2}$ and $\sigma_{2,1} = \tau \circ R_{2,1}$, where $\tau: V_i \otimes V_j \longrightarrow V_j \otimes V_i$ is the flip map, i.e. $\tau(v_1 \otimes v_2) = v_2 \otimes v_1$

$$\begin{array}{ccc}
\mathbb{C} & & \chi_1(g_2)\chi_2(g_1) \\
\uparrow \lambda_{\mathbb{C}} & & \uparrow \\
\mathbb{C} \otimes \mathbb{C} & & \chi_1(g_2) \otimes \chi_2(g_1) \\
\uparrow id_{\mathbb{C}} \otimes e'_{V_2} & & \uparrow \\
\mathbb{C} \otimes V_2 \otimes V_2^* & & \chi_1(g_2) \otimes \chi_2(g_1)e_2 \otimes e_2^* \\
\uparrow e_{V_1} \otimes id_{V_2} \otimes id_{V_2^*} & & \uparrow \\
V_1^* \otimes V_1 \otimes V_2 \otimes V_2^* & & e_1^* \otimes \chi_1(g_2)e_1 \otimes \chi_2(g_1)e_2 \otimes e_2^* \\
\uparrow id_{V_1^*} \otimes \sigma_{V_2, V_1} \otimes id_{V_2^*} & & \uparrow \\
V_1^* \otimes V_2 \otimes V_1 \otimes V_2^* & & e_1^* \otimes e_2 \otimes \chi_1(g_2)e_1 \otimes e_2^* \\
\uparrow id_{V_1^*} \otimes \sigma_{V_1, V_2} \otimes id_{V_2^*} & & \uparrow \\
V_1^* \otimes V_1 \otimes V_2 \otimes V_2^* & & e_1^* \otimes e_1 \otimes e_2 \otimes e_2^* \\
\uparrow id_{V_1^*} \otimes id_{V_1} \otimes i_{V_2} & & \uparrow \\
V_1^* \otimes V_1 \otimes \mathbb{C} & & e_1^* \otimes e_1 \otimes 1 \\
\uparrow i'_{V_1} \otimes id_{\mathbb{C}} & & \uparrow \\
\mathbb{C} \otimes \mathbb{C} & & 1 \otimes 1 \\
\uparrow \lambda_{\mathbb{C}}^{-1} & & \uparrow \\
\mathbb{C} & & 1
\end{array}$$

The scalar for this example is $\chi_1(g_2)\chi_2(g_1)$. The $n^2 \times n^2$ matrix is given by

$$(\chi_{i_1}(g_{j_2})\chi_{j_1}(g_{i_2}))_{0 \leq i_1, i_2, j_1, j_2 \leq n-1}$$

1.1. Abelian General Case. Here we show that the above formula generalizes when G is a finite abelian group. We consider the R -matrix

$$R = \sum_{g \in G} (1 \otimes g) \otimes (\delta_g \otimes e) = \sum_{g, u \in G} (\delta_u \otimes g) \otimes (\delta_g \otimes e),$$

We fix two representations (g_1, χ_1) and (g_2, χ_2) . The representation of R in $V_1 \otimes V_2$ is given by

$$\begin{aligned}
R_{1,2} &= \sum_{g, u \in G} \delta_{u, g_1} \chi_1(g) \otimes \delta_{g, g_2} \chi_2(e) \\
&= \sum_{u \in G} \delta_{u, g_1} \chi_1(g_2) \otimes 1 \\
&= \chi_1(g_2) \otimes 1
\end{aligned}$$

Similarly, the representation of R in $V_2 \otimes V_1$ is given by

$$R_{2,1} = \chi_2(g_1) \otimes 1$$

Now we consider a more complicated link with two components as shown in Figure 3.2. Here

$$R_{1,1} = \chi_1(g_1) \otimes 1, \quad R_{1,2} = \chi_1(g_2) \otimes 1, \quad R_{2,1} = \chi_2(g_1) \otimes 1, \quad R_{2,2} = \chi_2(g_2) \otimes 1$$

Similarly, the dual representation of (g, χ) is given by (g^{-1}, χ^{-1}) , therefore we get,

$$R_{1^*,2} = \chi_1^{-1}(g_2) \otimes 1, \quad R_{1,2^*} = \chi_1(g_2^{-1}) \otimes 1,$$

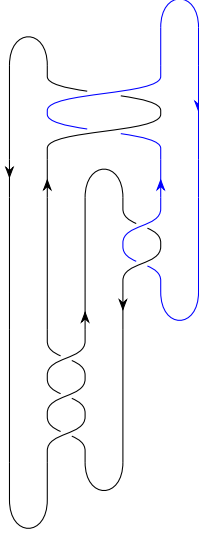


FIGURE 3.2. An example of a link with two components

We now compute the morphisms as in the case of the Hopf link, as shown in Figure 3.3.

The scalar for this link is

$$\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1})\chi_1^2(g_2)\chi_2^2(g_1) = \chi_1^3(g_1)\chi_1(g_2)\chi_2(g_1)$$

The $n^2 \times n^2$ matrix is given by

$$\left(\chi_{i_1}^3(g_{i_2})\chi_{i_1}(g_{j_2})\chi_{j_1}(g_{i_2}) \right)_{0 \leq i_1, i_2, j_1, j_2 \leq n-1}$$

where $n = |G|$ and each scalar value is determined by a choice of two elements g_{i_2} and g_{j_2} of G along with two characters χ_{i_1} and χ_{j_1} of G .

1.2. Abelian Case: General Formula for Links. Let G be a finite abelian group. In the case of a 2-component link, it is easy to note that each crossing contributes a scalar value, which can be factored out at each step. Moreover, after applying the evaluation and coevaluation maps, the basis elements eventually disappear, so the braiding contributes only a scalar. Therefore, in this case, knowing the signed (here signed is meant to make distinction between the crossings represented by σ and σ^{-1}) number of crossings of each type is enough to determine the resulting scalar.

We first note that a crossing of type $R_{a,b}$ contributes $\chi_a(g_b)$. Similarly, $R_{a^*,b}$ contributes $\chi_a^{-1}(g_b)$, R_{a,b^*} contributes $\chi_a(g_b^{-1})$, and R_{a^*,b^*} contributes $\chi_a^{-1}(g_b^{-1})$. The corresponding inverse braidings contribute the inverses of these scalars.

Thus, for a choice of two representations (g_1, χ_1) and (g_2, χ_2) , we obtain the following general formula for the scalar:

$$\prod_{\alpha, \beta} U_{\alpha, \beta}^{n_{\alpha, \beta}},$$

where the product runs over

$$\alpha, \beta \in \{1, 2, 1^*, 2^*\},$$

and

$$\begin{aligned} U_{a,b} &= \chi_a(g_b), & U_{a^*,b} &= \chi_a^{-1}(g_b), \\ U_{a,b^*} &= \chi_a(g_b^{-1}), & U_{a^*,b^*} &= \chi_a^{-1}(g_b^{-1}). \end{aligned}$$

Here,

$$n_{\alpha, \beta} = n_{\alpha, \beta}^+ - n_{\alpha, \beta}^-$$

$$\begin{array}{c}
\mathbb{C} \\
\lambda_{\mathbb{C}} \uparrow \\
\mathbb{C} \otimes \mathbb{C} \\
id_{\mathbb{C}} \otimes e'_{V_2} \uparrow \\
\mathbb{C} \otimes V_2 \otimes V_2^* \\
e_{V_1} \otimes id_{V_2 \otimes V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes V_2 \otimes V_2^* \\
id_{V_1^* \otimes \sigma_{V_2, V_1}} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_2 \otimes V_1 \otimes V_2^* \\
id_{V_1^* \otimes \sigma_{V_1, V_2}} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes V_2 \otimes V_2^* \\
id_{V_1^* \otimes \sigma_{V_2, V_1}} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_2 \otimes V_1 \otimes V_2^* \\
id_{V_1^* \otimes \sigma_{V_1, V_2}} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes V_2 \otimes V_2^* \\
id_{V_1^* \otimes V_1} \otimes \lambda_{V_2} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes \mathbb{C} \otimes V_2 \otimes V_2^* \\
id_{V_1^* \otimes V_1} \otimes e'_{V_1} \otimes id_{V_2 \otimes V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes V_1 \otimes V_1^* \otimes V_2 \otimes V_2^* \\
id_{V_1^* \otimes V_1 \otimes V_1} \otimes \sigma_{V_2, V_1^*} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes V_1 \otimes V_2 \otimes V_1^* \otimes V_2^* \\
id_{V_1^* \otimes V_1 \otimes V_1} \otimes \sigma_{V_1^*, V_2} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes V_1 \otimes V_1^* \otimes V_2 \otimes V_2^* \\
id_{V_1^* \otimes V_1 \otimes V_1 \otimes V_1^*} \otimes id_{V_2^*} \uparrow \\
V_1^* \otimes V_1 \otimes V_1 \otimes V_1^* \otimes \mathbb{C} \\
id_{V_1^* \otimes \sigma_{V_1, V_1}} \otimes id_{V_1^*} \otimes id_{\mathbb{C}} \uparrow \\
V_1^* \otimes V_1 \otimes V_1 \otimes V_1^* \otimes \mathbb{C} \\
id_{V_1^* \otimes \sigma_{V_1, V_1}} \otimes id_{V_1^*} \otimes id_{\mathbb{C}} \uparrow \\
V_1^* \otimes V_1 \otimes V_1 \otimes V_1^* \otimes \mathbb{C} \\
id_{V_1^* \otimes \sigma_{V_1, V_1}} \otimes id_{V_1^*} \otimes id_{\mathbb{C}} \uparrow \\
V_1^* \otimes V_1 \otimes V_1 \otimes V_1^* \otimes \mathbb{C} \\
id_{V_1^* \otimes V_1} \otimes i_{V_1} \otimes id_{\mathbb{C}} \uparrow \\
V_1^* \otimes V_1 \otimes \mathbb{C} \otimes \mathbb{C} \\
id_{V_1^* \otimes V_1} \otimes \lambda_{\mathbb{C}}^{-1} \uparrow \\
V_1^* \otimes V_1 \otimes \mathbb{C} \\
i'_{V_1} \otimes id_{\mathbb{C}} \uparrow \\
\mathbb{C} \otimes \mathbb{C} \\
\lambda_{\mathbb{C}}^{-1} \uparrow \\
\mathbb{C}
\end{array}
\quad
\begin{array}{c}
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1})\chi_1^2(g_2)\chi_2^2(g_1) \\
\uparrow \\
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1})\chi_1^2(g_2)\chi_2^2(g_1) \ (1 \otimes 1) \\
\uparrow \\
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1})\chi_1^2(g_2)\chi_2^2(g_1) \ (1 \otimes e_2 \otimes e_2^*) \\
\uparrow \\
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1}) \ (e_1^* \otimes \chi_1^2(g_2)e_1 \otimes \chi_2^2(g_1)e_2 \otimes e_2^*) \\
\uparrow \\
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1}) \ (e_1^* \otimes \chi_2(g_1)e_2 \otimes \chi_1^2(g_2)e_1 \otimes e_2^*) \\
\uparrow \\
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1}) \ (e_1^* \otimes \chi_1(g_2)e_1 \otimes \chi_2(g_1)e_2 \otimes e_2^*) \\
\uparrow \\
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1}) \ (e_1^* \otimes e_2 \otimes \chi_1(g_2)e_1 \otimes e_2^*) \\
\uparrow \\
\chi_1^3(g_1)\chi_1^{-1}(g_2)\chi_2(g_1^{-1}) \ (e_1^* \otimes e_1 \otimes e_2 \otimes e_2^*) \\
\uparrow \\
e_1^* \otimes \chi_1(g_1)e_1 \otimes \chi_1^2(g_1)\chi_1^{-1}(g_2) \otimes \chi_2(g_1^{-1})e_2 \otimes e_2^* \\
\uparrow \\
e_1^* \otimes \chi_1(g_1)e_1 \otimes \chi_1^2(g_1)e_1 \otimes \chi_1^{-1}(g_2)e_1^* \otimes \chi_2(g_1^{-1})e_2 \otimes e_2^* \\
\uparrow \\
e_1^* \otimes \chi_1(g_1)e_1 \otimes \chi_1^2(g_1)e_1 \otimes e_2 \otimes \chi_1^{-1}(g_2)e_1^* \otimes e_2^* \\
\uparrow \\
e_1^* \otimes \chi_1(g_1)e_1 \otimes \chi_1^2(g_1)e_1 \otimes e_1^* \otimes e_2 \otimes e_2^* \\
\uparrow \\
e_1^* \otimes \chi_1(g_1)e_1 \otimes \chi_1^2(g_1)e_1 \otimes e_1^* \otimes 1 \\
\uparrow \\
e_1^* \otimes \chi_1(g_1)e_1 \otimes \chi_1(g_1)e_1 \otimes e_1^* \otimes 1 \\
\uparrow \\
e_1^* \otimes e_1 \otimes \chi_1(g_1)e_1 \otimes e_1^* \otimes 1 \\
\uparrow \\
e_1^* \otimes e_1 \otimes e_1 \otimes e_1^* \otimes 1 \\
\uparrow \\
e_1^* \otimes e_1 \otimes 1 \otimes 1 \\
\uparrow \\
e_1^* \otimes e_1 \otimes 1 \\
\uparrow \\
1 \otimes 1 \\
\uparrow \\
1
\end{array}$$

FIGURE 3.3. Morphisms for the 2 component link in Fig 3.2

where $n_{\alpha,\beta}^+$ is the total number of positive crossings of type $R_{\alpha,\beta}$, and $n_{\alpha,\beta}^-$ is the total number of corresponding inverse crossings. Since, G is abelian, therefore χ is a homomorphism, and the formula further simplifies as follows:

$$\prod_{\alpha,\beta} U_{\alpha,\beta}^{n_{\alpha,\beta} + n_{\alpha^*,\beta^*} - n_{\alpha^*,\beta} - n_{\alpha,\beta^*}} = \prod_{\alpha,\beta} (\chi_\alpha(g_\beta))^{n_{\alpha,\beta} + n_{\alpha^*,\beta^*} - n_{\alpha^*,\beta} - n_{\alpha,\beta^*}},$$

where the product runs over

$$\alpha, \beta \in \{1, 2\}.$$

The formula also generalizes to a link with more components. The formula remains the same, except that the product is taken over

$$\alpha, \beta \in \{1, 2, \dots, r\}.$$

2. Representations of Drinfeld Doubles of Finite Groups

Throughout this section, we follow [3] for the classification and explicit construction of the irreducible representations of $D(G)$. The irreducible representations of $D(G)$ are classified by pairs (C_A, π) , where C_A is a conjugacy class of G and π is an irreducible representation of the centralizer of a fixed element of C_A .

Let $\{C_A\}_{A=0,\dots,p}$ be the set of conjugacy classes of G , in each C_A , we pick an element ${}^A g_1$ and consider the corresponding centralizer N_A . We choose a set of representatives ${}^A x_1, \dots, {}^A x_q$ of equivalence classes in G/N_A and we fix ${}^A x_1 = e$.

$$y \in N_A \iff y {}^A g_1 = {}^A g_1 y \iff y {}^A g_1 y^{-1} = {}^A g_1$$

Let $g \in C_A$, there exists $x \in G$ such that

$$x {}^A g_1 x^{-1} = g. \quad (11)$$

Since the cosets of N_A partition G , there exists some $1 \leq j \leq q$ and $h \in N_A$ such that $x = {}^A x_j h$. Now substituting this back in (11), we get

$$({}^A x_j h) {}^A g_1 ({}^A x_j h)^{-1} = g = {}^A x_j (h {}^A g_1 h^{-1}) {}^A x_j^{-1}.$$

Note that $h {}^A g_1 h^{-1} = {}^A g_1$. Conversely given an element ${}^A x_i {}^A g_1 {}^A x_i^{-1} \in C_A$, we get ${}^A x_i {}^A g_1 {}^A x_i^{-1} = {}^A g_j$. Which proves that

$$C_A = \left\{ {}^A x_1 {}^A g_1 {}^A x_1^{-1} = {}^A g_1, {}^A x_2 {}^A g_1 {}^A x_2^{-1} = {}^A g_2, \dots, {}^A x_q {}^A g_1 {}^A x_q^{-1} = {}^A g_q \right\}.$$

Let $\pi : N_A \rightarrow \text{End}(V_\pi)$ be an irreducible representation. Define the vector space $D(G) \otimes_{B_A} V_\pi$, where B_A is the subalgebra of $D(G)$ spanned by elements of the form $\delta_h \otimes x$, where h is an arbitrary element of G while x is an arbitrary element of N_A . Define representation ρ_π of B_A on V_π as follows:

$$\rho_\pi(\delta_h \otimes x) = \delta_{h, {}^A g_1} \pi(x)$$

This representation of B_A induces a representation on the vector space $D(G) \otimes_{B_A} V_\pi$.

Then a basis of the induced vector space $W_\pi^A = D(G) \otimes_{B_A} V_\pi$ is given by the set of vectors:

$$\{|{}^A x_j, e_i\rangle = (1 \otimes {}^A x_j) \otimes_{B_A} e_i\}$$

where:

- $\{{}^A x_1, \dots, {}^A x_q\}$ is a set of representatives of the left cosets G/N_A , with ${}^A x_1 = e$. These elements map the chosen representative ${}^A g_1$ to each element in the conjugacy class via ${}^A g_j = {}^A x_j {}^A g_1 {}^A x_j^{-1}$.

- $\{e_1, \dots, e_{\dim V_\pi}\}$ is the chosen basis for the internal vector space V_π .
- 1 denotes the unit element of the function algebra $\text{Map}(G, \mathbb{C})$, specifically $1 = \sum_{h \in G} \delta_h$.

Define representation ρ_π^A of $D(G)$ on W_π^A as follows:

$$\rho_\pi^A(\delta_{\tilde{g}} \otimes x)(|{}^A x_j, e_i\rangle) = \delta_{\tilde{g}, x^A g_j x^{-1}} |{}^A x_k, \pi(h) e_i\rangle, \quad (12)$$

where ${}^A x_k$ and h are defined by the relation $x^A x_j = {}^A x_k h$ and $h \in N_A$.

3. Dual Representations

Given a representation $\rho_\pi^A : D(G) \longrightarrow \text{End}(W_\pi^A)$ as before, we define the dual representation

$$(\rho_\pi^A)^* : D(G) \longrightarrow \text{End}(W_\pi^{A*}).$$

The dual basis of W_π^A is given by $\{\langle {}^A x^j, e^i | \rangle\}$ with the relation $\langle {}^A x^k, e^l | \rangle |{}^A x_j, e_i\rangle = \delta_j^k \delta_i^l$. We recall that the antipode for the Drinfeld double is given by

$$S(\delta_g \otimes x) = \delta_{x^{-1} g^{-1} x} \otimes x^{-1}.$$

The action of a basis element $\delta_g \otimes x$ on $(W_\pi^A)^*$ via the dual representation is defined as follows: Given an element $f \in W_\pi^{A*}$

$$(\rho^*(\delta_g \otimes x)f)(v) = f(\rho(S(\delta_g \otimes x))v),$$

for all $v \in W_\pi^A$, where $S : D(G) \rightarrow D(G)$ is the antipode of the Hopf algebra $D(G)$.

This is the standard dual action for finite-dimensional modules over a Hopf algebra; see [6, Section 5.2].

The antipode serves the role of the inverse element in the group case. It satisfies the antihomomorphism property

$$S(\alpha\beta) = S(\beta)S(\alpha),$$

which ensures that the above action defines a valid representation of $D(G)$.

In our setting, the action on a basis element of the dual W_π^{A*} is given by

$$(\rho_\pi^{A*}(\delta_g \otimes x)f)(v) = f(\rho_\pi^A(\delta_{x^{-1} g^{-1} x} \otimes x^{-1})v).$$

Taking $f = \langle {}^A x^j, e^i | \rangle$ and $v = |{}^A x_p, e_q\rangle$ and applying the definition of ρ_π^A , we get,

$$\begin{aligned} (\rho_\pi^{A*}(\delta_g \otimes x)\langle {}^A x^j, e^i | \rangle) |{}^A x_p, e_q\rangle &= \langle {}^A x^j, e^i | \delta_{x^{-1} g^{-1} x, x^{-1} A_{g_p} x} |{}^A x_k, \pi(h) e_q\rangle. \\ &= \delta_{g^{-1}, A_{g_p}} \langle {}^A x^j, e^i | |{}^A x_k, \pi(h) e_q\rangle. \end{aligned} \quad (13)$$

where, $x^{-1} A_{g_p} x = {}^A x_k h$.

Since $\pi(h)$ is a matrix, let

$$\pi(h) = (\pi(h)_{r,s})_{1 \leq r,s \leq \dim V_\pi}.$$

Therefore,

$$\begin{aligned} \pi(h) e_q &= \sum_{r=1}^{\dim V_\pi} \pi(h)_{r,q} e_r \\ \langle {}^A x^j, e^i | |{}^A x_k, \pi(h) e_q\rangle &= \langle {}^A x^j, e^i | |{}^A x_k, \sum_{r=1}^{\dim V_\pi} \pi(h)_{r,q} e_r\rangle \\ &= \sum_{r=1}^{\dim V_\pi} \pi(h)_{r,q} \langle {}^A x^j, e^i | |{}^A x_k, e_r\rangle \end{aligned} \quad (14)$$

where ${}^A x_k$ and h are defined by the relation $x^{-1} {}^A x_p = {}^A x_k h$ and $h \in N_A$.

Now substituting (14) in (13), we get,

$$\left(\rho_\pi^{A*}(\delta_g \otimes x) \langle {}^A x^j, e^i | \right) | {}^A x_p, e_q \rangle = \delta_{g^{-1}, {}^A g_p} \sum_{r=1}^{\dim V_\pi} \pi(h)_{r,q} \langle {}^A x^j, e^i | | {}^A x_k, e_r \rangle, \quad (15)$$

$$= \delta_{g^{-1}, {}^A g_p} \sum_{r=1}^{\dim V_\pi} \pi(h)_{r,q} \delta_{j,k} \delta_{i,r}, \quad \text{using orthogonality.} \quad (16)$$

where ${}^A x_k$ and h are defined by the relation $x^{-1} {}^A x_p = {}^A x_k h$ and $h \in N_A$.

Further, to clarify the notation, we write $h = h(p)$ and $k = k(p)$, meaning that when g and x are fixed, then h and k depend only on p . We get the following:

$$\left(\rho_\pi^{A*}(\delta_g \otimes x) \langle {}^A x^j, e^i | \right) | {}^A x_p, e_q \rangle = \delta_{g^{-1}, {}^A g_p} \sum_{r=1}^{\dim V_\pi} \pi(h(p))_{r,q} \delta_{j,k(p)} \delta_{i,r}.$$

Now, using the fact that the dual basis is determined by its action on the basis elements of W_π^A , we can reconstruct the dual vector in the image vector space. For a linear functional $\Psi \in W_\pi^{A*}$, we have

$$\Psi = \sum_{p,q} \Psi(| {}^A x_p, e_q \rangle) \langle {}^A x^p, e^q |.$$

Therefore, substituting the expression obtained above, we get

$$\begin{aligned} \left(\rho_\pi^{A*}(\delta_g \otimes x) \langle {}^A x^j, e^i | \right) &= \sum_{p,q} \delta_{g^{-1}, {}^A g_p} \sum_{r=1}^{\dim V_\pi} \pi(h(p))_{r,q} \delta_{j,k(p)} \delta_{i,r} \langle {}^A x^p, e^q | \\ &= \sum_{p,q} \delta_{g^{-1}, {}^A g_p} \delta_{j,k(p)} \pi(h(p))_{i,q} \langle {}^A x^p, e^q |. \end{aligned} \quad (17)$$

The condition

$$\delta_{j,k(p)}$$

restricts the sum to those p for which

$$x^{-1} {}^A x_p = {}^A x_j h(p).$$

Equivalently, multiplying by x from the left, we have

$${}^A x_p = x {}^A x_j h(p).$$

Since $\{{}^A x_p\}$ is a set of representatives of the cosets of G/N_A , there exists a unique index k and an element $h \in N_A$ such that

$$x {}^A x_j = {}^A x_k h^{-1}.$$

Thus, only the term $p = k$ contributes to the sum in (17).

Renaming g as $\tilde{g}$ for clarity, the action of the dual representation on a basis element is given by

$$\rho_\pi^{A*}(\delta_{\tilde{g}} \otimes x) \langle {}^A x^j, e^i | = \delta_{\tilde{g}^{-1}, {}^A g_k} \sum_{q=1}^{\dim V_\pi} \pi(h)_{i,q} \langle {}^A x^k, e^q |. \quad (18)$$

where k and $h \in N_A$ are determined by the relation

$$x {}^A x_j = {}^A x_k h^{-1}.$$

4. The Universal R-Matrix

Given two representations (C_{A_1}, π_1) and (C_{A_2}, π_2) , we will denote different representations of the R-matrix as follows:

$$\begin{aligned} R_{1,2} &: W_{\pi_1}^{A_1} \otimes W_{\pi_2}^{A_2} \longrightarrow W_{\pi_1}^{A_1} \otimes W_{\pi_2}^{A_2}, \\ R_{1^*,2} &: \left(W_{\pi_1}^{A_1}\right)^* \otimes W_{\pi_2}^{A_2} \longrightarrow \left(W_{\pi_1}^{A_1}\right)^* \otimes W_{\pi_2}^{A_2}, \\ R_{1,2^*} &: W_{\pi_1}^{A_1} \otimes \left(W_{\pi_2}^{A_2}\right)^* \longrightarrow W_{\pi_1}^{A_1} \otimes \left(W_{\pi_2}^{A_2}\right)^*, \\ R_{1^*,2^*} &: \left(W_{\pi_1}^{A_1}\right)^* \otimes \left(W_{\pi_2}^{A_2}\right)^* \longrightarrow \left(W_{\pi_1}^{A_1}\right)^* \otimes \left(W_{\pi_2}^{A_2}\right)^* \end{aligned}$$

Now we want to compute the R-matrix corresponding to irreducible representations given by (C_{A_1}, π_1) and (C_{A_2}, π_2) . In our definition,

$$R = \sum_{g \in G} (1 \otimes g) \otimes (\delta_g \otimes e).$$

Note that $1 = \sum_{u \in G} \delta_u$ Therefore,

$$R = \sum_{u, g \in G} (\delta_u \otimes g) \otimes (\delta_g \otimes e). \quad (19)$$

Now, we compute the action of the operators $\rho_\pi^A(\delta_u \otimes g)$, $\rho_\pi^A(\delta_g \otimes e)$, $(\rho_\pi^A)^*(\delta_u \otimes g)$, and $(\rho_\pi^A)^*(\delta_g \otimes e)$ on the basis elements of W_π^A and $(W_\pi^A)^*$, respectively.

Using (12) with $\tilde{g} = u$ and $x = g$, we obtain

$$\rho_\pi^A(\delta_u \otimes g)|^A x_j, e_i\rangle = \delta_{u, g^A g_j g^{-1}}|^A x_k, \pi(h)e_i\rangle, \quad (20)$$

where $^A x_k$ and h are defined by the relation $g^A x_j = ^A x_k h$ and $h \in N_A$.

Similarly, using (12) with $\tilde{g} = g$ and $x = e$, we obtain

$$\rho_\pi^A(\delta_g \otimes e)|^A x_j, e_i\rangle = \delta_{g, ^A g_j}|^A x_k, \pi(h)e_i\rangle,$$

where $^A x_k$ and h are defined by the relation $^A x_j = ^A x_k h$ and $h \in N_A$.

In this case, $j = k$ and $h = e$, therefore we get

$$\rho_\pi^A(\delta_g \otimes e)|^A x_j, e_i\rangle = \delta_{g, ^A g_j}|^A x_j, e_i\rangle, \quad (21)$$

Using (18) with $\tilde{g} = u$ and $x = g$, we obtain

$$\rho_\pi^{A^*}(\delta_u \otimes g)\langle^A x^j, e^i| = \delta_{u^{-1}, ^A g_k} \sum_{q=1}^{\dim V_\pi} \pi(h)_{i,q} \langle^A x^k, e^q|, \quad (22)$$

where k and $h \in N_A$ are determined by the relation

$$g^A x_j = ^A x_k h^{-1}.$$

Similarly, using (18) with $\tilde{g} = g$ and $x = e$, we obtain

$$\rho_\pi^{A^*}(\delta_g \otimes e)\langle^A x^j, e^i| = \delta_{g^{-1}, ^A g_k} \sum_{q=1}^{\dim V_\pi} \pi(h)_{i,q} \langle^A x^k, e^q|,$$

where k and $h \in N_A$ are determined by the relation

$$^A x_j = ^A x_k h^{-1}.$$

In this case $k = j$ and $h = e$, therefore, we get:

$$\rho_{\pi}^{A*}(\delta_g \otimes e) \langle A x^j, e^i | = \delta_{g^{-1}, A g_j} \langle A x^j, e^i |. \quad (23)$$

4.1. Action of $R_{1,2}$. Now we see how $R_{1,2}$ acts on a basis element $|^{A_1} x_{j_1}, e_{i_1}\rangle \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle$ of $W_{\pi_1}^{A_1} \otimes W_{\pi_2}^{A_2}$. Using (19), (20) and (21), we get,

$$\begin{aligned} R_{1,2} |^{A_1} x_{j_1}, e_{i_1}\rangle \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle &= \sum_{u, g \in G} \delta_{u, g^{A_1} g_{j_1} g^{-1}} |^{A_1} x_{k_1}, \pi_1(h_1) e_{i_1}\rangle \otimes \delta_{g, A_2 g_{j_2}} |^{A_2} x_{j_2}, e_{i_2}\rangle, \\ &= \sum_{u \in G} \delta_{u, A_2 g_{j_2}^{A_1} g_{j_1} (A_2 g_{j_2})^{-1}} |^{A_1} x_{k_1}, \pi_1(h_1) e_{i_1}\rangle \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle, \\ &= |^{A_1} x_{k_1}, \pi_1(h_1) e_{i_1}\rangle \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle, \end{aligned} \quad (24)$$

where $^{A_1} x_{k_1}$ and h_1 are defined by the relation $A_2 g_{j_2}^{A_1} x_{j_1} = ^{A_1} x_{k_1} h_1$ and $h_1 \in N_{A_1}$.

4.2. Action of $R_{1*,2}$. Now we see how $R_{1*,2}$ acts on a basis element $\langle ^{A_1} x^{j_1}, e^{i_1} | \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle$ of $(W_{\pi_1}^{A_1})^* \otimes W_{\pi_2}^{A_2}$.

Using (19), (22) and (21), we get,

$$\begin{aligned} R_{1*,2} \langle ^{A_1} x^{j_1}, e^{i_1} | \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle &= \sum_{u, g \in G} \left(\delta_{u^{-1}, A_1 g_{k_1}} \sum_{q=1}^{\dim V_{\pi_1}} \pi_1(h_1)_{i_1, q} \langle ^{A_1} x^{k_1}, e^q | \right) \otimes \delta_{g, A_2 g_{j_2}} |^{A_2} x_{j_2}, e_{i_2}\rangle, \\ &= \sum_{u \in G} \left(\delta_{u^{-1}, A_1 g_{k_1}} \sum_{q=1}^{\dim V_{\pi_1}} \pi_1(h_1)_{i_1, q} \langle ^{A_1} x^{k_1}, e^q | \right) \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle, \\ &= \sum_{q=1}^{\dim V_{\pi_1}} \pi_1(h_1)_{i_1, q} \langle ^{A_1} x^{k_1}, e^q | \otimes |^{A_2} x_{j_2}, e_{i_2}\rangle, \end{aligned} \quad (25)$$

where k_1 and $h_1 \in N_{A_1}$ are determined by the relation

$$A_2 g_{j_2}^{A_1} x_{j_1} = ^{A_1} x_{k_1} h_1^{-1}.$$

4.3. Action of $R_{1,2*}$. Now we see how $R_{1,2*}$ acts on a basis element $|^{A_1} x_{j_1}, e_{i_1}\rangle \otimes \langle ^{A_2} x^{j_2}, e^{i_2} |$ of $W_{\pi_1}^{A_1} \otimes (W_{\pi_2}^{A_2})^*$. Using (19), (20) and (23), we get,

$$\begin{aligned} R_{1,2*} |^{A_1} x_{j_1}, e_{i_1}\rangle \otimes \langle ^{A_2} x^{j_2}, e^{i_2} | &= \sum_{u, g \in G} \delta_{u, g^{A_1} g_{j_1} g^{-1}} |^{A_1} x_{k_1}, \pi_1(h_1) e_{i_1}\rangle \otimes \delta_{g^{-1}, A_2 g_{j_2}} \langle ^{A_2} x^{j_2}, e^{i_2} |, \\ &= \sum_{u \in G} \delta_{u, (A_2 g_{j_2})^{-1} A_1 g_{j_1} A_2 g_{j_2}} |^{A_1} x_{k_1}, \pi_1(h_1) e_{i_1}\rangle \otimes \langle ^{A_2} x^{j_2}, e^{i_2} |, \\ &= |^{A_1} x_{k_1}, \pi_1(h_1) e_{i_1}\rangle \otimes \langle ^{A_2} x^{j_2}, e^{i_2} |, \end{aligned} \quad (26)$$

where $^{A_1} x_{k_1}$ and h_1 are defined by the relation $(A_2 g_{j_2})^{-1} A_1 x_{j_1} = ^{A_1} x_{k_1} h_1$ and $h_1 \in N_{A_1}$.

4.4. Action of $R_{1^*,2^*}$. Now we see how $R_{1^*,2^*}$ acts on a basis element $\langle^{A_1}x^{j_1}, e^{i_1} | \otimes \langle^{A_2}x^{j_2}, e^{i_2} |$ of $(W_{\pi_1}^{A_1})^* \otimes (W_{\pi_2}^{A_2})^*$. Using (19), (22) and (23), we get,

$$\begin{aligned} R_{1^*,2^*} \langle^{A_1}x^{j_1}, e^{i_1} | \otimes \langle^{A_2}x^{j_2}, e^{i_2} | &= \sum_{u,g \in G} \left(\delta_{u^{-1}, A_1 g_{k_1}} \sum_{q=1}^{\dim V_{\pi_1}} \pi_1(h_1)_{i_1,q} \langle^{A_1}x^{k_1}, e^q | \right) \otimes \delta_{g^{-1}, A_2 g_{j_2}} \langle^{A_2}x^{j_2}, e^{i_2} |, \\ &= \sum_{u \in G} \left(\delta_{u^{-1}, A_1 g_{k_1}} \sum_{q=1}^{\dim V_{\pi_1}} \pi_1(h_1)_{i_1,q} \langle^{A_1}x^{k_1}, e^q | \right) \otimes \langle^{A_2}x^{j_2}, e^{i_2} |, \\ &= \sum_{q=1}^{\dim V_{\pi_1}} \pi_1(h_1)_{i_1,q} \langle^{A_1}x^{k_1}, e^q | \otimes \langle^{A_2}x^{j_2}, e^{i_2} |, \end{aligned} \quad (27)$$

where k_1 and $h_1 \in N_{A_1}$ are determined by the relation

$$(A_2 g_{j_2})^{-1} A_1 x_{j_1} = A_1 x_{k_1} h_1^{-1}.$$

5. The Hopf-Link Invariant in the Non-Abelian Case

As an example, we consider the Hopf link shown in Figure 3.1. Let the two representations be given by the pairs

$$(C_{A_1}, \pi_1) \quad \text{and} \quad (C_{A_2}, \pi_2).$$

The corresponding morphism $\mathbb{C} \rightarrow \mathbb{C}$ is given by the following composition of maps.

$$\begin{aligned} \lambda_{\mathbb{C}} \circ (\text{id}_{\mathbb{C}} \otimes e'_{V_2}) \circ (e_{V_1} \otimes \text{id}_{V_2} \otimes \text{id}_{V_2^*}) \\ \circ (\text{id}_{V_1^*} \otimes \sigma_{V_2, V_1} \otimes \text{id}_{V_2^*}) \circ (\text{id}_{V_1^*} \otimes \sigma_{V_1, V_2} \otimes \text{id}_{V_2^*}) \\ \circ (\text{id}_{V_1^* \otimes V_1} \otimes i_{V_2}) \circ (i'_{V_1} \otimes \text{id}_{\mathbb{C}}) \circ \lambda_{\mathbb{C}}^{-1}. \end{aligned}$$

Using the composition property of tensor products, this can be written as

$$\begin{aligned} \lambda_{\mathbb{C}} \circ (\text{id}_{\mathbb{C}} \otimes e'_{V_2}) \circ (e_{V_1} \otimes \text{id}_{V_2} \otimes \text{id}_{V_2^*}) \\ \circ \left(\text{id}_{V_1^*} \otimes (\sigma_{V_2, V_1} \circ \sigma_{V_1, V_2}) \otimes \text{id}_{V_2^*} \right) \\ \circ (\text{id}_{V_1^* \otimes V_1} \otimes i_{V_2}) \circ (i'_{V_1} \otimes \text{id}_{\mathbb{C}}) \circ \lambda_{\mathbb{C}}^{-1}. \end{aligned} \quad (28)$$

To avoid computing long equations repeatedly, we first compute the image of 1 under the following map

$$(\text{id}_{V_1^* \otimes V_1} \otimes i_{V_2}) \circ (i'_{V_1} \otimes \text{id}_{\mathbb{C}}) \circ \lambda_{\mathbb{C}}^{-1}(1) = (\text{id}_{V_1^* \otimes V_1} \otimes i_{V_2}) \circ (i'_{V_1} \otimes \text{id}_{\mathbb{C}})(1 \otimes 1),$$

$$\begin{aligned} &= (\text{id}_{V_1^* \otimes V_1} \otimes i_{V_2}) \left(\sum_{i_1, j_1} \langle^{A_1}x^{j_1}, e^{i_1} | \otimes |^{A_1}x_{j_1}, e_{i_1} \rangle \right) \otimes 1, \\ &= \left(\sum_{i_1, j_1} \langle^{A_1}x^{j_1}, e^{i_1} | \otimes |^{A_1}x_{j_1}, e_{i_1} \rangle \right) \otimes \left(\sum_{i_2, j_2} |^{A_2}x_{j_2}, e_{i_2} \rangle \otimes \langle^{A_2}x^{j_2}, e^{i_2} | \right), \\ &= \sum_{i_1, j_1, i_2, j_2} \langle^{A_1}x^{j_1}, e^{i_1} | \otimes |^{A_1}x_{j_1}, e_{i_1} \rangle \otimes |^{A_2}x_{j_2}, e_{i_2} \rangle \otimes \langle^{A_2}x^{j_2}, e^{i_2} |. \end{aligned} \quad (29)$$

Further, we compute the following. Here, the braiding morphisms σ_{V_1, V_2} and σ_{V_2, V_1} are obtained from the corresponding R -matrices $R_{1,2}$ and $R_{2,1}$, respectively, by composing with the

appropriate flip maps.

$$\begin{aligned}
& \left(\text{id}_{V_1^*} \otimes (\sigma_{V_2, V_1} \circ \sigma_{V_1, V_2}) \otimes \text{id}_{V_2^*} \right) \left(\langle {}^{A_1}x^{j_1}, e^{i_1} | \otimes | {}^{A_1}x_{j_1}, e_{i_1} \rangle \otimes | {}^{A_2}x_{j_2}, e_{i_2} \rangle \otimes \langle {}^{A_2}x^{j_2}, e^{i_2} | \right) \\
&= \langle {}^{A_1}x^{j_1}, e^{i_1} | \otimes (\sigma_{V_2, V_1} \circ \sigma_{V_1, V_2}) \left(| {}^{A_1}x_{j_1}, e_{i_1} \rangle \otimes | {}^{A_2}x_{j_2}, e_{i_2} \rangle \right) \otimes \langle {}^{A_2}x^{j_2}, e^{i_2} |, \\
&= \langle {}^{A_1}x^{j_1}, e^{i_1} | \otimes \sigma_{V_2, V_1} \left(| {}^{A_2}x_{j_2}, e_{i_2} \rangle \otimes | {}^{A_1}x_{k_1}, \pi_1(h_1)e_{i_1} \rangle \right) \otimes \langle {}^{A_2}x^{j_2}, e^{i_2} |, \\
&= \left(\langle {}^{A_1}x^{j_1}, e^{i_1} | \otimes | {}^{A_1}x_{k_1}, \pi_1(h_1)e_{i_1} \rangle \right) \otimes \left(| {}^{A_2}x_{k_2}, \pi_2(h_2)e_{i_2} \rangle \otimes \langle {}^{A_2}x^{j_2}, e^{i_2} | \right). \tag{30}
\end{aligned}$$

where ${}^{A_1}x_{k_1}$, ${}^{A_2}x_{k_2}$ and h_1, h_2 are defined by the relations

$${}^{A_2}g_{j_2} {}^{A_1}x_{j_1} = {}^{A_1}x_{k_1} h_1, \quad {}^{A_1}g_{k_1} {}^{A_2}x_{j_2} = {}^{A_2}x_{k_2} h_2,$$

and $h_1 \in N_{A_1}$ and $h_2 \in N_{A_2}$.

Now, using (29) and (30) with (28) we get the following

$$\begin{aligned}
& (\text{id}_{V_1^*} \otimes (\sigma_{V_2, V_1} \circ \sigma_{V_1, V_2}) \otimes \text{id}_{V_2^*}) \circ (\text{id}_{V_1^* \otimes V_1} \otimes i_{V_2}) \circ (i'_{V_1} \otimes \text{id}_{\mathbb{C}}) \circ \lambda_{\mathbb{C}}^{-1}(1) \\
&= \sum_{i_1, j_1, i_2, j_2} \left(\langle {}^{A_1}x^{j_1}, e^{i_1} | \otimes | {}^{A_1}x_{k_1}, \pi_1(h_1)e_{i_1} \rangle \right) \otimes \left(| {}^{A_2}x_{k_2}, \pi_2(h_2)e_{i_2} \rangle \otimes \langle {}^{A_2}x^{j_2}, e^{i_2} | \right), \\
&= \sum_{i_1, j_1, i_2, j_2} \left(\langle {}^{A_1}x^{j_1}, e^{i_1} | \otimes | {}^{A_1}x_{k_1}, \pi_1(h_1)e_{i_1} \rangle \right) \otimes \left(| {}^{A_2}x_{k_2}, \pi_2(h_2)e_{i_2} \rangle \otimes \langle {}^{A_2}x^{j_2}, e^{i_2} | \right), \\
&= \sum_{i_1, j_1, i_2, j_2} \delta_{j_1, k_1} \delta_{j_2, k_2} e^{i_1}(\pi_1(h_1)e_{i_1}) e^{i_2}(\pi_2(h_2)e_{i_2}), \\
&= \sum_{\substack{j_1, j_2 \\ j_1=k_1, j_2=k_2}} \text{Tr}(\pi_1(h_1)) \text{Tr}(\pi_2(h_2)), \\
&= \sum_{\substack{j_1, j_2 \\ j_1=k_1, j_2=k_2}} \chi_1(h_1) \chi_2(h_2),
\end{aligned}$$

where ${}^{A_1}x_{k_1}$, ${}^{A_2}x_{k_2}$ and h_1, h_2 are defined by the relations ${}^{A_2}g_{j_2} {}^{A_1}x_{j_1} = {}^{A_1}x_{k_1} h_1$ and ${}^{A_1}g_{k_1} {}^{A_2}x_{j_2} = {}^{A_2}x_{k_2} h_2$ and $h_1 \in N_{A_1}$ and $h_2 \in N_{A_2}$.

The relations simplify as follows: $h_1 = ({}^{A_1}x_{k_1})^{-1} {}^{A_2}g_{j_2} {}^{A_1}x_{j_1} \in N_{A_1}$, similarly $h_2 = ({}^{A_2}x_{k_2})^{-1} {}^{A_1}g_{j_1} {}^{A_2}x_{j_2} \in N_{A_2}$.

Therefore, we get

$$\sum \chi_1(h_1) \chi_2(h_2),$$

where the sum runs over all pairs $(h_1, h_2) \in N_{A_1} \times N_{A_2}$ arising from the braiding relations

$$h_1 = ({}^{A_1}x_{j_1})^{-1} {}^{A_2}g_{j_2} {}^{A_1}x_{j_1}, \tag{31}$$

$$h_2 = ({}^{A_2}x_{j_2})^{-1} {}^{A_1}g_{j_1} {}^{A_2}x_{j_2}. \tag{32}$$

6. The S_3 Example and Its Unnormalized S-Matrix

The symmetric group S_3 has three conjugacy classes. The choice of representative of each conjugacy class, corresponding centralizer N_i , and the choice of representatives of classes in the quotient S_3/N_i are summarized in Table 3.1.

Now we write solutions to (31) and (32) for different choices of $A_1, A_2 \in \{0, 1, 2\}$.

TABLE 3.1. Conjugacy classes, centralizers, and coset representatives of S_3

Class	${}^i g_1$	N_i	S_3/N_i
0	$\{{}^0 g_1 = e\}$	S_3	$\{{}^0 x_1 = e\}$
1	$\{{}^1 g_1 = (12), {}^1 g_2 = (23), {}^1 g_3 = (13)\}$	$\{e, (12)\}$	$\{{}^1 x_1 = e, {}^1 x_2 = (13), {}^1 x_3 = (23)\}$
2	$\{{}^2 g_1 = (123), {}^2 g_2 = (132)\}$	$\{e, (123), (132)\}$	$\{{}^2 x_1 = e, {}^2 x_2 = (12)\}$

Case $A_1 = 0, A_2 = 0$: The only solution to (31) and (32) is

$$(h_1, h_2) = (e, e).$$

Case $A_1 = 0, A_2 \neq 0$: The solutions to (31) and (32) are

$$(h_1, h_2) = ({}^{A_2} g_{j_2}, e),$$

where ${}^{A_2} g_{j_2}$ is an arbitrary element of the conjugacy class C_{A_2} .

Case $A_1 \neq 0, A_2 = 0$: The solutions to (31) and (32) are

$$(h_1, h_2) = (e, {}^{A_1} g_{j_1}),$$

where ${}^{A_1} g_{j_1}$ is an arbitrary element of the conjugacy class C_{A_1} .

Case $A_1 = A_2 = 1$: Since any representative of a class in S_3/N_i as chosen in Table 3.1 is either identity or a transposition, therefore, ${}^A x_i^{-1} = {}^A x_i$. Consequently, (31) and (32) simplify to

$$h_1 = {}^1 x_{j_1} {}^1 g_{j_2} {}^1 x_{j_1}, \quad h_2 = {}^1 x_{j_2} {}^1 g_{j_1} {}^1 x_{j_2}.$$

Now, we compute possible values of h_1 .

$$\begin{aligned}
h_1 &= {}^1 x_1 {}^1 g_1 {}^1 x_1 = (12) \in N_1 \\
h_1 &= {}^1 x_1 {}^1 g_2 {}^1 x_1 = (23) \notin N_1 \\
h_1 &= {}^1 x_1 {}^1 g_3 {}^1 x_1 = (13) \notin N_1 \\
h_1 &= {}^1 x_2 {}^1 g_1 {}^1 x_2 = (13)(12)(13) = (23) \notin N_1 \\
h_1 &= {}^1 x_2 {}^1 g_2 {}^1 x_2 = (13)(23)(13) = (12) \in N_1 \\
h_1 &= {}^1 x_2 {}^1 g_3 {}^1 x_2 = (13)(13)(13) = (13) \notin N_1 \\
h_1 &= {}^1 x_3 {}^1 g_1 {}^1 x_3 = (23)(12)(23) = (13) \notin N_1 \\
h_1 &= {}^1 x_3 {}^1 g_2 {}^1 x_3 = (23)(23)(23) = (23) \notin N_1 \\
h_1 &= {}^1 x_3 {}^1 g_3 {}^1 x_3 = (23)(13)(23) = (12) \in N_1
\end{aligned}$$

For the second relation, we consider only the cases where $h_1 \in N_1$.

$$\begin{aligned}
h_2 &= {}^1 x_1 {}^1 g_1 {}^1 x_1 = (12) \in N_1 \\
h_2 &= {}^1 x_2 {}^1 g_2 {}^1 x_2 = (13)(23)(13) = (12) \in N_1 \\
h_2 &= {}^1 x_3 {}^1 g_3 {}^1 x_3 = (23)(13)(23) = (12) \in N_1
\end{aligned}$$

The only pair satisfying the relation is $(h_1, h_2) = ((12), (12))$ with multiplicity three.

Now, when $A_1 = A_2 = 2$, we get the following relations for h_1 .

$$h_1 = {}^2x_1 {}^2g_1 {}^2x_1 = (123) \in N_2$$

$$h_1 = {}^2x_1 {}^2g_2 {}^2x_1 = (132) \in N_2$$

$$h_1 = {}^2x_2 {}^2g_1 {}^2x_2 = (12)(123)(12) = (132) \in N_2$$

$$h_1 = {}^2x_2 {}^2g_2 {}^2x_2 = (12)(132)(12) = (123) \in N_2$$

The relations for h_2

$$h_2 = {}^2x_1 {}^2g_1 {}^2x_1 = (123) \in N_2$$

$$h_2 = {}^2x_2 {}^2g_1 {}^2x_2 = (12)(123)(12) = (132) \in N_2$$

$$h_2 = {}^2x_1 {}^2g_2 {}^2x_1 = (132) \in N_2$$

$$h_2 = {}^2x_2 {}^2g_2 {}^2x_2 = (12)(132)(12) = (123) \in N_2$$

So the surviving pairs (h_1, h_2) are $((123), (123)), ((132), (132))$, each with multiplicity two.

Now for $A_1 = 1$ and $A_2 = 2$, the relations for h_1 ,

$$h_1 = {}^1x_1 {}^2g_1 {}^1x_1 = (123) \notin N_1$$

$$h_1 = {}^1x_1 {}^2g_2 {}^1x_1 = (132) \notin N_1$$

$$h_1 = {}^1x_3 {}^2g_1 {}^1x_3 = (23)(123)(23) = (132) \notin N_1$$

$$h_1 = {}^1x_3 {}^2g_2 {}^1x_3 = (23)(132)(23) = (123) \notin N_1$$

$$h_1 = {}^1x_2 {}^2g_1 {}^1x_2 = (13)(123)(13) = (132) \notin N_1$$

$$h_1 = {}^1x_2 {}^2g_2 {}^1x_2 = (13)(132)(13) = (123) \notin N_1$$

In all these cases $h_1 \notin N_1$, therefore when $A_1 = 1$ and $A_2 = 2$ (similarly $A_2 = 1$ and $A_1 = 2$, because in this case $h_2 \notin N_1$) there are no solutions of the equations (31) and (32).

After imposing the braiding relations (31) and (32), only certain pairs $(h_1, h_2) \in N_{A_1} \times N_{A_2}$ contribute to the sum. The surviving pairs and their multiplicities are summarized below:

Therefore, the Hopf-link sum reduces in each case to a sum over the surviving pairs listed above. In particular, the mixed sectors $(A_1, A_2) = (1, 2)$ and $(2, 1)$ give no contribution, since the braiding relations do not admit any solutions.

We use the standard character tables of S_3 , $N_1 \cong \mathbb{Z}_2$, and $N_2 \cong \mathbb{Z}_3$; see, for example, [8, Section 2.5, p. 20, and Section 5.1]. There are three irreducible representations of S_3 . Let $\chi_1^0, \chi_2^0, \chi_3^0$ denote the corresponding characters of the trivial, sign, and standard representations, respectively. Their values are given by

S_3	e	(12)	(123)
χ_1^0	1	1	1
χ_2^0	1	-1	1
χ_3^0	2	0	-1

(A_1, A_2)	Surviving (h_1, h_2)	Multiplicity
$(0, 0)$	(e, e)	1
$(0, 1)$	$({}^1g_j, e), \quad j = 1, 2, 3$	1 for each j
$(1, 0)$	$(e, {}^1g_j), \quad j = 1, 2, 3$	1 for each j
$(0, 2)$	$({}^2g_j, e), \quad j = 1, 2$	1 for each j
$(2, 0)$	$(e, {}^2g_j), \quad j = 1, 2$	1 for each j
$(1, 1)$	$((12), (12))$	3
$(2, 2)$	$((123), (123))$	2
	$((132), (132))$	2
$(1, 2)$	none	0
$(2, 1)$	none	0

TABLE 3.2. Surviving pairs (h_1, h_2) obtained from the braiding relations for different choices of conjugacy classes (A_1, A_2) in S_3 .

Similarly, N_1 has two irreducible representations. Denoting their characters by χ_1^1 and χ_2^1 , corresponding to the trivial and nontrivial representations, respectively, we have

N_1	e	(12)
χ_1^1	1	1
χ_2^1	1	-1

Finally, N_2 has three irreducible representations. Let $\chi_1^2, \chi_2^2, \chi_3^2$ denote the corresponding characters, where χ_1^2 is the trivial representation and χ_2^2, χ_3^2 are the two nontrivial one-dimensional representations. Writing

$$\omega = e^{2\pi i/3},$$

their values are

N_2	e	(123)	(132)
χ_1^2	1	1	1
χ_2^2	1	ω	ω^2
χ_3^2	1	ω^2	ω

where $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$.

The S -matrix for this example is an 8×8 matrix. To simplify the computation, we first compute blocks of the matrix, $S_{0,0}, S_{0,1}, \dots, S_{2,2}$, where $S_{m,n}$ gives values of the sum $\sum \chi_1(h_1) \chi_2(h_2)$ in the case when χ_1 is an irreducible character of N_m and χ_2 is an irreducible character of N_n with $m, n \in \{0, 1, 2\}$. Also, using Table 3.2, we get the following values for each block of S -matrix.

For $S_{0,0}$, we get the formula $\chi_m^0(e) \chi_n^0(e)$, which is the product of dimensions of irreducible representations that correspond to χ_m^0 and χ_n^0 . Therefore, we get

$$S_{0,0} = \begin{array}{c|ccc} & \chi_1^0 & \chi_2^0 & \chi_3^0 \\ \hline \chi_1^0 & 1 & 1 & 2 \\ \chi_2^0 & 1 & 1 & 2 \\ \chi_3^0 & 2 & 2 & 4 \end{array}$$

For the block $S_{0,1}$ the entries are given as follows:

$$\sum \chi_m^0(h_1) \chi_n^1(h_2) = \sum_{j=1}^3 \chi_m^0({}^1g_j) \chi_n^1(e) = \sum_{j=1}^3 \chi_m^0({}^1g_j)$$

because each irreducible representation of $\mathbb{Z}_2 \cong N_1$ is one-dimensional. Since all 1g_j 's are conjugates of each other, this further simplifies the sum as follows:

$$\sum_{j=1}^3 \chi_m^0({}^1g_j) = 3\chi_m^0((12))$$

Therefore, we get

$$S_{0,1} = \begin{array}{c|cc} & \chi_1^1 & \chi_2^1 \\ \hline \chi_1^0 & 3\chi_1^0((12)) & 3\chi_1^0((12)) \\ \chi_2^0 & 3\chi_2^0((12)) & 3\chi_2^0((12)) \\ \chi_3^0 & 3\chi_3^0((12)) & 3\chi_3^0((12)) \end{array} = \begin{array}{c|cc} & \chi_1^1 & \chi_2^1 \\ \hline \chi_1^0 & 3 & 3 \\ \chi_2^0 & -3 & -3 \\ \chi_3^0 & 0 & 0 \end{array}$$

We use similar reduction in $S_{0,2}$, $S_{1,0}$ and $S_{2,0}$.

Similarly, we get

$$\sum \chi_m^0(h_1) \chi_n^2(h_2) = 2\chi_m^0(123)$$

$$S_{0,2} = \begin{array}{c|ccc} & \chi_1^2 & \chi_2^2 & \chi_3^2 \\ \hline \chi_1^0 & 2\chi_1^0(123) & 2\chi_1^0(123) & 2\chi_1^0(123) \\ \chi_2^0 & 2\chi_2^0(123) & 2\chi_2^0(123) & 2\chi_2^0(123) \\ \chi_3^0 & 2\chi_3^0(123) & 2\chi_3^0(123) & 2\chi_3^0(123) \end{array} = \begin{array}{c|ccc} & \chi_1^2 & \chi_2^2 & \chi_3^2 \\ \hline \chi_1^0 & 2 & 2 & 2 \\ \chi_2^0 & 2 & 2 & 2 \\ \chi_3^0 & -2 & -2 & -2 \end{array}$$

We also note that $S_{2,0}$ and $S_{1,0}$ are the transposes of $S_{0,2}$ and $S_{0,1}$ respectively.

$$S_{1,1} = \begin{array}{c|cc} & \chi_1^1 & \chi_2^1 \\ \hline \chi_1^1 & 3\chi_1^1((12))\chi_1^1((12)) & 3\chi_1^1((12))\chi_2^1((12)) \\ \chi_2^1 & 3\chi_2^1((12))\chi_1^1((12)) & 3\chi_2^1((12))\chi_2^1((12)) \end{array} = \begin{array}{c|cc} & \chi_1^1 & \chi_2^1 \\ \hline \chi_1^1 & 3 & -3 \\ \chi_2^1 & -3 & 3 \end{array}$$

$$S_{1,2} = 0 = \begin{array}{c|ccc} & \chi_1^2 & \chi_2^2 & \chi_3^2 \\ \hline \chi_1^1 & 0 & 0 & 0 \\ \chi_2^1 & 0 & 0 & 0 \end{array}$$

$$S_{2,1} = 0 = \begin{array}{c|cc} & \chi_1^1 & \chi_2^1 \\ \hline \chi_1^2 & 0 & 0 \\ \chi_2^2 & 0 & 0 \\ \chi_3^2 & 0 & 0 \end{array}$$

Let $a = (123)$ and $b = (132)$. We get the following table

$$S_{2,2} = \begin{array}{c|ccc} & \chi_1^2 & \chi_2^2 & \chi_3^2 \\ \hline \chi_1^2 & 2\chi_1^2(a)\chi_1^2(a) + 2\chi_1^2(b)\chi_1^2(b) & 2\chi_1^2(a)\chi_2^2(a) + 2\chi_1^2(b)\chi_2^2(b) & 2\chi_1^2(a)\chi_3^2(a) + 2\chi_1^2(b)\chi_3^2(b) \\ \chi_2^2 & 2\chi_2^2(a)\chi_1^2(a) + 2\chi_2^2(b)\chi_1^2(b) & 2\chi_2^2(a)\chi_2^2(a) + 2\chi_2^2(b)\chi_2^2(b) & 2\chi_2^2(a)\chi_3^2(a) + 2\chi_2^2(b)\chi_3^2(b) \\ \chi_3^2 & 2\chi_3^2(a)\chi_1^2(a) + 2\chi_3^2(b)\chi_1^2(b) & 2\chi_3^2(a)\chi_2^2(a) + 2\chi_3^2(b)\chi_2^2(b) & 2\chi_3^2(a)\chi_3^2(a) + 2\chi_3^2(b)\chi_3^2(b) \end{array}$$

which, on further simplification, gives

$$\begin{array}{c|ccc} & \chi_1^2 & \chi_2^2 & \chi_3^2 \\ \hline \chi_1^2 & 4 & 2(\omega^2 + \omega) & 2(\omega^2 + \omega) \\ \chi_2^2 & 2(\omega^2 + \omega) & 2(\omega^2 + \omega) & 4 \\ \chi_3^2 & 2(\omega^2 + \omega) & 4 & 2(\omega^2 + \omega) \end{array} = \begin{array}{c|ccc} & \chi_1^2 & \chi_2^2 & \chi_3^2 \\ \hline \chi_1^2 & 4 & -2 & -2 \\ \chi_2^2 & -2 & -2 & 4 \\ \chi_3^2 & -2 & 4 & -2 \end{array}$$

With the irreducible representations ordered according to the three conjugacy-class sectors, the resulting unnormalized S -matrix has the following block decomposition. Under the normalization convention in which the matrix is divided by $|S_3| = 6$, the normalized matrix is $\frac{1}{6}S$.

$$S = \left[\begin{array}{c|c|c} S_{0,0} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix} & S_{0,1} = \begin{pmatrix} 3 & 3 \\ -3 & -3 \\ 0 & 0 \end{pmatrix} & S_{0,2} = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ -2 & -2 & -2 \end{pmatrix} \\ \hline S_{1,0} = \begin{pmatrix} 3 & -3 & 0 \\ 3 & -3 & 0 \end{pmatrix} & S_{1,1} = \begin{pmatrix} 3 & -3 \\ -3 & 3 \end{pmatrix} & S_{1,2} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \hline S_{2,0} = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ 2 & 2 & -2 \end{pmatrix} & S_{2,1} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} & S_{2,2} = \begin{pmatrix} 4 & -2 & -2 \\ -2 & -2 & 4 \\ -2 & 4 & -2 \end{pmatrix} \end{array} \right]$$

Conclusion

In this thesis, we studied link invariants associated with the Drinfeld doubles of finite groups. We first introduced the categorical structures needed to describe framed tangles by ribbon diagrams. We then considered the Drinfeld doubles of finite cyclic groups. In this case, the irreducible representations are one-dimensional, which allowed us to derive an explicit formula for the unnormalized S -matrix of the Hopf link. We also used the same representation-theoretic method to compute the invariant of another two-component link.

For a general finite group, we described the irreducible representations of $D(G)$ in terms of conjugacy classes and irreducible representations of their centralizers. We computed the corresponding dual representations and derived the actions of the universal R -matrix on representations and their duals. Using these formulas, we obtained an expression for the Hopf-link invariant as a sum involving characters of the relevant centralizers.

Finally, we applied this expression to S_3 . By determining the contributing pairs of centralizer elements and using their character tables, we calculated the complete unnormalized S -matrix. After division by $|S_3| = 6$, this gives the normalized S -matrix under the convention used in this thesis.

Future Directions

There are several possible directions in which these computations can be extended. The same method can be applied to other non-abelian finite groups, such as dihedral groups or symmetric groups of higher degree. This would provide further examples of how the conjugacy classes and centralizer representations determine the entries of the S -matrix.

It would also be interesting to apply the R -matrix formulas to links with more components or more crossings. Since such calculations become longer as the group and the link become more complicated, they could be implemented in a computer algebra system.

A further direction would be to study link invariants arising from more general quantum groups. In particular, one could replace the Drinfeld doubles of finite groups by quantum groups such as $U_q(\mathfrak{sl}_2)$ and investigate how their representations and universal R -matrices produce link invariants. This would extend the methods used here to a broader setting in which the representation theory and the corresponding R -matrices are more complicated.

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